

Web Appendices for “Doubly Robust Proximal Causal Inference under Confounded Outcome-Dependent Sampling”

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A Proofs of Theorems.

A.1 Proof of Proposition 1

To prove Equation (2), we have

$$\begin{aligned}
& E[q(A, Z, X) \mid A, W, X, Y = 0, S = 1] \\
&= E\{E[q(A, Z, X) \mid A, U, W, X, Y = 0, S = 1] \mid A, W, X, Y = 0, S = 1\} \\
&\stackrel{A.2}{=} E\{E[q(A, Z, X) \mid A, U, X, Y = 0, S = 1] \mid A, W, X, Y = 0, S = 1\} \\
&\stackrel{A.3}{=} E\left\{\frac{1}{P(A \mid U, X, Y = 0, S = 1)} \mid A, W, X, Y = 0, S = 1\right\} \\
&= \int \frac{1}{P(A \mid U = u, X, Y = 0, S = 1)} f(u \mid A, W, X, Y = 0, S = 1) du \\
&= \int \frac{1}{P(A \mid U = u, X, Y = 0, S = 1)} \times \frac{f(u \mid W, X, Y = 0, S = 1)P(A \mid U = u, W, X, Y = 0, S = 1)}{P(A \mid W, X, Y = 0, S = 1)} du \tag{S1}
\end{aligned}$$

$$\begin{aligned}
&\stackrel{A.2}{=} \int \frac{1}{P(A \mid U = u, X, Y = 0, S = 1)} \times \frac{f(u \mid W, X, Y = 0, S = 1)P(A \mid U = u, X, Y = 0, S = 1)}{P(A \mid W, X, Y = 0, S = 1)} du \tag{S2} \\
&= \frac{1}{P(A \mid W, X, Y = 0, S = 1)} \int f(u \mid W, X, Y = 0, S = 1) du \\
&= \frac{1}{P(A \mid W, X, Y = 0, S = 1)}.
\end{aligned}$$

To prove the uniqueness of a solution to Equation (1) under Assumption 4(b), we suppose there exists another function q^* that solves Equation (2), that is,

$$E\{q^*(A, Z, X) \mid A, W, X, Y = 0, S = 1\} = 1/P(A \mid W, X, Y = 0, S = 1).$$

Then

$$E\{q^*(A, Z, X) - q(A, Z, X) \mid A, W, X, Y = 0, S = 1\} = 0.$$

By Assumption 4(b), $q^*(A, Z, X) = q(A, Z, X)$ almost surely.

Suppose there exists a function q_0 that solves Equation (1), then q_0 is also the solution to (2) by the proof above. By the uniqueness of a solution to Equation (2), $q_0(A, Z, X) = q(A, Z, X)$ almost surely, which

proves that q is the unique solution to Equation (1).

A.2 Proof of Proposition 2

We first state a useful result for identification:

Lemma S1. *Under Model 1 and a selection mechanism that satisfies Assumption 1, the log odds ratio satisfies*

$$\beta_0 = \log(\theta_{1c}/\theta_{0c}), \quad (\text{S3})$$

where, for $a = 0, 1$,

$$\theta_{ac} = E \left\{ c(X) \frac{P(A = a \mid U, X, Y = 1, S = 1)}{P(A = a \mid U, X, Y = 0, S = 1)} P(Y = 1 \mid U, X, S = 1) \mid S = 1 \right\}. \quad (\text{S4})$$

Proof. Under Model 1 and Assumption 1, we have

$$\begin{aligned} & \frac{P(Y = 1 \mid U, X, A = 1, S = 1)P(Y = 0 \mid U, X, A = 0, S = 1)}{P(Y = 1 \mid U, X, A = 0, S = 1)P(Y = 0 \mid U, X, A = 1, S = 1)} \\ &= \exp(\beta_0) \times \frac{P(S = 1 \mid Y = 1, A = 1, U, X)}{P(S = 1 \mid Y = 0, A = 1, U, X)} \times \frac{P(S = 1 \mid Y = 0, A = 0, U, X)}{P(S = 1 \mid Y = 1, A = 0, U, X)} \\ &\stackrel{A.1}{=} \exp(\beta_0) \times \xi(U, X) \times \xi(U, X)^{-1} = \exp(\beta_0). \end{aligned}$$

Therefore

$$\begin{aligned} \theta_{1c} &= E \left[c(X) \frac{P(A = 1 \mid U, X, Y = 1, S = 1)}{P(A = 1 \mid U, X, Y = 0, S = 1)} P(Y = 1 \mid U, X, S = 1) \mid S = 1 \right] \\ &= E \left[c(X) \frac{P(A = 0 \mid U, X, Y = 1, S = 1)}{P(A = 0 \mid U, X, Y = 0, S = 1)} P(Y = 1 \mid U, X, S = 1) \mid S = 1 \right] \exp(\beta_0) \\ &= \theta_{0c} \exp(\beta_0), \end{aligned}$$

which proves Lemma S1. □

By Lemma S1, it suffices to prove that $\theta_{ac} = E[c(X)I(A = a)Yq(A, Z, X) \mid S = 1]$. Under Assumptions 2 and 4, $q(A, Z, X)$ solves Equation (1). We have

$$\begin{aligned}
& E\{F_{\text{PIPW}}(a, Y, Z, X; c, q) \mid S = 1\} \tag{S5} \\
& = E[c(X)I(A = a)Yq(a, Z, X) \mid S = 1] \\
& = E[c(X)I(A = a)YE(q(a, Z, X) \mid A = a, U, X, Y = 1, S = 1) \mid S = 1] \\
& \stackrel{A.2}{=} E[c(X)I(A = a)YE(q(a, Z, X) \mid A = a, U, X, Y = 0, S = 1)Y \mid S = 1] \\
& \stackrel{A.3}{=} E[c(X)I(A = a)Y \frac{1}{P(A = a \mid U, X, Y = 0, S = 1)} \mid S = 1] \\
& = E[c(X)E\{I(A = a) \mid U, X, Y = 1, S = 1\} \frac{1}{P(A = a \mid U, X, Y = 0, S = 1)} Y \mid S = 1] \\
& = E[c(X) \frac{P(A = a \mid U, X, Y = 1, S = 1)}{P(A = a \mid U, X, Y = 0, S = 1)} Y \mid S = 1] \\
& = E[c(X) \frac{P(A = a \mid U, X, Y = 1, S = 1)}{P(A = a \mid U, X, Y = 0, S = 1)} P(Y = 1 \mid U, X, S = 1) \mid S = 1] \\
& = \theta_{ac}. \tag{S6}
\end{aligned}$$

A.3 Proof of Theorem 1

First, We show that Equation (5) is equivalent to the moment equation

$$E\{(1 - Y)h(a, W, X) \mid A = a, U, X, S = 1\} = E(Y \mid A = a, U, X, S = 1). \quad (\text{S7})$$

This is because

$$\begin{aligned} & E\{(1 - Y)h(a, W, X) \mid A = a, U, X, S = 1\} \\ &= E\{h(a, W, X) \mid A = a, U, X, Y = 0, S = 1\}P(Y = 0 \mid A = a, U, X, S = 1) \\ &\stackrel{\text{A.5}}{=} \frac{P(Y = 1 \mid A = a, U, X, S = 1)}{P(Y = 0 \mid A = a, U, X, S = 1)}P(Y = 0 \mid A = a, U, X, S = 1) \\ &= P(Y = 1 \mid A = a, U, X, S = 1) \\ &= E(Y \mid A = a, U, X, S = 1). \end{aligned}$$

Next, we show that Assumption 2 and Equation (5) implies the moment equation

$$E\{(1 - Y)h(a, W, X) \mid A = a, Z, X, S = 1\} = E(Y \mid A = a, Z, X, S = 1). \quad (\text{S8})$$

This is because

$$\begin{aligned} & E\{(1 - Y)h(A, W, X) - Y \mid A, Z, X, S = 1\} \\ &= E[E\{(1 - Y)h(A, W, X) - Y \mid A, U, Z, X, S = 1\} \mid A, Z, X, S = 1] \\ &\stackrel{\text{A.2}}{=} E[E\{(1 - Y)h(A, W, X) - Y \mid A, U, X, S = 1\} \mid A, Z, X, S = 1] \\ &\stackrel{\text{Eq(S7)}}{=} 0. \end{aligned}$$

Finally, the left-hand side of Equation (S8) equals

$$E\{h(a, W, X) \mid A = a, Z, X, Y = 0, S = 1\}P(Y = 0 \mid A = a, Z, X, S = 1).$$

Rearranging the terms, we obtain Equation (5).

To prove the uniqueness of a solution to Equation (5) under Assumption 6(b), we suppose there exists

another function h^* that solves Equation (5), that is,

$$E\{h^*(a, W, X) \mid A = a, Z, X, Y = 0, S = 1\} = \frac{P(Y = 1 \mid A = a, Z, X, S = 1)}{P(Y = 0 \mid A = a, Z, X, S = 1)}$$

Then

$$E\{h^*(a, W, X) - h(a, W, X) \mid A = a, Z, X, Y = 0, S = 1\} = 0.$$

By Assumption 6(b), $h^*(A, Z, X) = h(A, Z, X)$ almost surely.

Suppose there exists a function h_0 that solves Equation (5), then h_0 is also the solution to Equation (5) by the proof above. By the uniqueness of a solution to Equation (5), $h_0(A, Z, X) = h(A, Z, X)$ almost surely, which proves that h is the unique solution to Equation (5).

A.4 Proof of Theorem 2

Under Model 1 and Assumptions 1 and 2, the result in Lemma S1 holds. It suffices to prove that

$$\theta_{ac} = E\{c(X)(1 - Y)h(a, W, X) \mid S = 1\} \quad (\text{S9})$$

for $a = 0, 1$.

For $a = 0, 1$, we have that

$$\begin{aligned} & E[c(X)(1 - Y)h(a, W, X) \mid S = 1] \\ &= E[c(X)(1 - Y)E\{h(a, W, X) \mid U, X, Y = 0, S = 1\} \mid S = 1] \\ &\stackrel{\text{A.2}}{=} E[c(X)(1 - Y)E\{h(a, W, X) \mid A = a, U, X, Y = 0, S = 1\} \mid S = 1] \\ &\stackrel{\text{A.5}}{=} E[c(X)(1 - Y)\frac{P(Y = 1 \mid A = a, U, X, S = 1)}{P(Y = 0 \mid A = a, U, X, S = 1)} \mid S = 1] \\ &\stackrel{\text{Eq. (4)}}{=} \theta_{ac}. \end{aligned}$$

A.5 Proof of Theorem 3

We first find an influence function $IF(\beta_0; q, h, c)$ by finding a pathwise derivative. Let $h(A, Z, X)$ be the outcome confounding bridge function defined as the solution to Equation (5), or equivalently, Equation (S8). Consider a regular one-dimensional parametric submodel of $\mathcal{M}_{sp}$ indexed by t , where $t = 0$ indicates the truth. We write h_t as the solution to Equation (S8) at t under the parametric submodel.

Also denote E_t and f_t respectively as the expectation operator and density function under the parametric submodel. By abuse of notation, we write $R(\cdot)$ as the score function at $t = 0$, i.e. $R(\cdot) = \partial \log f_t(\cdot) / \partial t |_{t=0}$, and write

$$R(O) = \frac{\partial}{\partial t} \log f_t(A, Y, Z, W, X, S) |_{t=0}$$

We write β_{0t} as the solution to Equation (6) under the parametric submodel, that is, β_{0t} solves

$$E_t [c(X)(1 - Y)\{h(1, W, X)e^{-\beta_{0t}} - h(0, W, X)\} | S = 1] = 0,$$

or equivalently,

$$E_t [Sc(X)(1 - Y)\{h(1, W, X)e^{-\beta_{0t}} - h(0, W, X)\}] = 0. \quad (\text{S10})$$

We denote $\dot{\beta}_0$ and $\dot{h}(A, W, X)$ as the pathwise derivative of β_{0t} and $h(A, W, X)$ evaluated at $t = 0$, respectively.

Taking derivative with respect to t and evaluating at $t = 0$ on both sides of Equation (S10), we have

$$\begin{aligned} 0 &= \frac{\partial}{\partial t} E_t [c(X)(1 - Y)\{h(1, W, X)e^{-\beta_{0t}} - h(0, W, X)\} | S = 1] |_{t=0} \\ &= E [Sc(X)(1 - Y)\{h(1, W, X)e^{-\beta_0} - h(0, W, X)\}R(O)] + \\ &\quad E [Sc(X)(1 - Y)\{\dot{h}(1, W, X)e^{-\beta_0} - h(c, W, X)\}] - \\ &\quad E \{Sc(X)(1 - Y)h(1, W, X)e^{-\beta_0}\} \dot{\beta}_0 \end{aligned}$$

Below we calculate $E[Sc(X)(1 - Y)\dot{h}(1, W, X)e^{-\beta_0}]$ and $E[Sc(X)(1 - Y)\dot{h}(0, W, X)]$, by Equation (S8), the function h_t satisfies

$$0 = E_t [\{(1 - Y)h_t(A, W, X) - Y\} | A, Z, X, S = 1].$$

Differentiating both sides with respect to t and evaluating at $t = 0$, we obtain

$$\begin{aligned} E[\{Y - (1 - Y)h(A, W, X)\} R(Y, W \mid A, Z, X, S = 1) \mid A, Z, X, S = 1] = \\ E\left\{(1 - Y)\dot{h}(A, W, X) - Y \mid A, Z, X, S = 1\right\} \end{aligned} \quad (\text{S11})$$

After rearranging the terms, Equation (S11) implies

$$E[S(1 - Y)\dot{h}(A, W, X)n(A, Z, X)] \quad (\text{S12})$$

$$= E[S\{Y - (1 - Y)h(A, W, X)\}n(A, Z, X)R(O)] + \quad (\text{S13})$$

$$E[SYn(A, Z, X)] \quad (\text{S14})$$

Above, taking $n(A, Z, X) = c(X)Ae^{-\beta_0 A}q(A, Z, X)$. Because the function q satisfies Equation (2), we have

$$\begin{aligned} (\text{S12}) &= E\left[Sc(X)(1 - Y)\dot{h}(A, W, X)Ae^{-\beta_0 A}q(A, Z, X)\right] \\ &= E\left[Sc(X)(1 - Y)\dot{h}(A, W, X)Ae^{-\beta_0 A}E\{q(A, Z, X) \mid A, W, X, Y = 0, S = 1\}\right] \\ &\stackrel{Eq(2)}{=} E\left[Sc(X)(1 - Y)\dot{h}(A, W, X)Ae^{-\beta_0 A}\frac{1}{f(A \mid W, X, Y = 0, S = 1)}\right] \\ &= E\left[Sc(X)(1 - Y)E\left\{A\dot{h}(A, W, X)e^{-\beta_0 A} \times \right. \right. \\ &\quad \left. \left. \frac{1}{f(A \mid W, X, Y = 0, S = 1)} \mid W, X, Y = 0, S = 1\right\}\right] \\ &= E\left\{Sc(X)(1 - Y)\dot{h}(1, W, X)e^{-\beta_0}\right\}, \\ (\text{S13}) &= E\left[Sc(X)\{Y - (1 - Y)h(A, W, X)\}Ae^{-\beta_0 A}q(A, Z, X)R(O)\right], \\ (\text{S14}) &= E\left[Sc(X)AYe^{-\beta_0 A}q(A, Z, X)\right]. \end{aligned}$$

Similarly, taking $n(A, Z, X) = c(X)(1 - A)q(A, Z, X)$, we have

$$(S12) = E \left[Sc(X)(1 - Y)\dot{h}(A, W, X)(1 - A)q(A, Z, X) \right]$$

$$= E \left\{ Sc(X)(1 - Y)\dot{h}(0, W, X) \right\},$$

$$(S13) = E [Sc(X)\{Y - (1 - Y)h(A, W, X)\}(1 - A)q(A, Z, X)R(O)],$$

$$(S14) = E [Sc(X)(1 - A)Yq(A, Z, X)].$$

We therefore have

$$\begin{aligned} & E \left[Sc(X)(1 - Y)\{\dot{h}(1, W, X)e^{-\beta_0} - \dot{h}(0, W, X)\} \right] \\ &= E \left[Sc(X)\{Y - (1 - Y)\dot{h}(A, W, X)\}(-1)^{1-A}e^{-\beta_0 A}q(A, Z, X)R(O) \right] + \\ & \quad E \{ Sc(X)(-1)^{1-A}e^{-\beta_0 A}Yq(A, Z, X) \} \\ & \stackrel{T.2}{=} E \left[Sc(X)\{Y - (1 - Y)\dot{h}(A, W, X)\}(-1)^{1-A}e^{-\beta_0 A}q(A, Z, X)R(O) \right] \end{aligned}$$

Hence

$$\begin{aligned} \dot{\beta}_0 &= [E \{ Sc(X)AYq(A, Z, X)e^{-\beta_0 A} \}]^{-1} \times \\ & \quad E \left\{ Sc(X) \left[(-1)^{1-A}Yq(A, Z, X)e^{-\beta_0 A} - \right. \right. \\ & \quad \left. \left. (-1)^{1-A}(1 - Y)q(A, Z, X)h(A, W, X)e^{-\beta_0 A} + \right. \right. \\ & \quad \left. \left. (1 - Y) \{ h(1, W, X)e^{-\beta_0} - h(0, W, X) \} \right] R(O) \right\} \\ &= [E \{ Sc(X)AYq(A, Z, X)e^{-\beta_0 A} \}]^{-1} \times \\ & \quad E \left\{ Sc(X) \left[(-1)^{1-A}q(A, Z, X)e^{-\beta_0 A} \{ Y - (1 - Y)h(A, W, X) \} + \right. \right. \\ & \quad \left. \left. (1 - Y) \{ h(1, W, X)e^{-\beta_0} - h(0, W, X) \} \right] R(O) \right\} \end{aligned}$$

and therefore an influence function for β_0 is

$$\begin{aligned}
IF(\beta_0; q, h, c) &= [E\{Sc(X)h(1, W, X)e^{-\beta_0}\}]^{-1} \times \\
&\quad Sc(X) \left[(-1)^{1-A} q(A, Z, X) e^{-\beta_0 A} \{Y - (1 - Y)h(A, W, X)\} + \right. \\
&\quad \left. (1 - Y) \{h(1, W, X)e^{-\beta_0} - h(0, W, X)\} \right] \\
&= \frac{1}{\eta_c} \left\{ Sc(X) \left[(-1)^{1-A} q(A, Z, X) e^{-\beta_0 A} \{Y - (1 - Y)h(A, W, X)\} + \right. \right. \\
&\quad \left. \left. (1 - Y) \{h(1, W, X)e^{-\beta_0} - h(0, W, X)\} \right] \right\}.
\end{aligned}$$

Now we characterize the optimal index c^* that yields the influence function with minimum variance. The variance of $IF(\beta_0; q, h, c)$ is

$$\begin{aligned}
&E[\{IF(\beta_0; q, h, c)\}^2] \\
&= \frac{E[\{Sc(X)M(O; \beta_0, q, h)\}^2]}{[E\{Sc(X)(1 - Y)h(1, W, X)e^{-\beta_0}\}]^2} \\
&= \frac{E[\{c(X)\}^2 E\{M(O; \beta_0, q, h)\}^2 \mid X, S = 1 \mid S = 1]}{\{E[c(X)E\{(1 - Y)h(1, W, X) \mid X, S = 1\} \mid S = 1]\}^2 e^{-2\beta_0} P(S = 1)} \\
&= \frac{E[\{c(X)\}^2 Var\{M(O; \beta_0, q, h) \mid X, S = 1\} \mid S = 1]}{\{E[c(X)E\{(1 - Y)h(1, W, X) \mid X, S = 1\} \mid S = 1]\}^2 e^{-2\beta_0} P(S = 1)}
\end{aligned}$$

where

$$\begin{aligned}
M(O; q, h, \beta_0) &= (-1)^{1-A} q(A, Z, X) e^{-\beta_0 A} \{Y - (1 - Y)h(A, W, X)\} + \\
&\quad (1 - Y) \{h(1, W, X)e^{-\beta_0} - h(0, W, X)\}.
\end{aligned}$$

By Cauchy-Shwarz inequality, we have

$$\begin{aligned}
&E[\{c(X)\}^2 Var\{M(O; \beta_0, q, h) \mid X, S = 1\} \mid S = 1] \times \\
&E\left\{\frac{[E\{(1 - Y)h(1, W, X) \mid X, S = 1\}]^2}{Var\{M(O; \beta_0, q, h) \mid X, S = 1\}} \mid S = 1\right\} \geq \\
&\{E[c(X)E\{(1 - Y)h(1, W, X) \mid X, S = 1\} \mid S = 1]\}^2
\end{aligned}$$

and therefore

$$E[\{IF(\beta_0; q, h, c)\}^2] \geq \left[E \left\{ \frac{[E\{(1-Y)h(1, W, X) \mid X, S=1\}]^2}{Var\{M(O; \beta_0, q, h) \mid X, S=1\}} \mid S=1 \right\} e^{-2\beta_0} P(S=1) \right]^{-1}.$$

The equality holds if and only if

$$c(X) = \frac{\alpha^* E\{(1-Y)h(1, W, X) \mid X, S=1\}}{Var\{M(O; \beta_0, q, h) \mid X, S=1\}}$$

for any real-valued $\alpha^* \neq 0$. This gives the EIF among all β_0 's that has the smallest variance, $IF(\beta_{c^*}; q, h, c^*)$.

A.6 Proof of Theorem 4

To prove

$$\begin{aligned} \beta_0 = \log \bigg(& E \left\{ c(X) \left[I(A=1)q(A, Z, X) \{Y - (1-Y)h(A, W, X)\} + \right. \right. \\ & \left. \left. (1-Y)h(1, W, X) \right] \mid S=1 \right\} / E \left\{ c(X) \left[I(A=1)q(A, Z, X) \{Y - \right. \right. \\ & \left. \left. (1-Y)h(A, W, X)\} + (1-Y)h(0, W, X) \right] \mid S=1 \right\} \bigg), \end{aligned}$$

by Lemma S1 it suffices to prove

$$\begin{aligned} \theta_{ac} = E \{ c(X) [(1-Y)h(a, W, X) - \\ I(A=a)q(a, Z, X) \{ (1-Y)h(A, W, X) - Y \}] \mid S=1 \}. \end{aligned} \quad (\text{S15})$$

for $a = 0, 1$.

(a) If q solves Equation (2), then the right-hand side of Equation (S15) equals

$$\begin{aligned}
& E \left\{ c(X) [(1-Y)h(a, W, X) - I(A=a)(1-Y)q(a, Z, X)h(A, W, X) + \right. \\
& \quad \left. I(A=a)q(a, Z, X)Y] \mid S=1 \right\} \\
& \stackrel{Eq(S6)}{=} E \left\{ c(X)(1-Y)h(a, W, X) [1 - I(A=a)q(a, Z, X)] \mid S=1 \right\} + \theta_{ac} \\
& = E \left\{ c(X)(1-Y)h(a, W, X) E[1 - \right. \\
& \quad \left. I(A=a)q(a, Z, X) \mid A, W, X, Y=0, S=1] \mid S=1 \right\} + \theta_{ac} \\
& \stackrel{Eq(2)}{=} E \left\{ c(X)(1-Y)h(a, W, X) \left[1 - \frac{I(A=a)}{P(A=a \mid W, X, Y=0, S=1)} \right] \mid S=1 \right\} + \theta_{ac} \\
& = E \left\{ c(X)(1-Y) \left[h(a, W, X) - \frac{I(A=a)h(a, W, X)}{P(A=a \mid W, X, Y=0, S=1)} \right] \mid S=1 \right\} + \theta_{ac} \\
& = E \left\{ c(X)(1-Y) E[h(a, W, X) - \right. \\
& \quad \left. \frac{I(A=a)h(a, W, X)}{P(A=a \mid W, X, Y=0, S=1)} \mid W, X, Y=0, S=1] \mid S=1 \right\} + \theta_{ac} \\
& = E \left\{ c(X)(1-Y) [h(a, W, X) - h(a, W, X)] \mid S=1 \right\} + \theta_{ac} \\
& = \theta_{ac}.
\end{aligned}$$

(b) If h solves Equation (5), or equivalently, Equation (S8), then the right-hand side of Equation (S15) equals

$$\begin{aligned}
& E \left\{ c(X) \left[(1-Y)h(a, W, X) - I(A=a)(1-Y)q(a, Z, X)h(a, W, X) + \right. \right. \\
& \quad \left. \left. I(A=a)q(a, Z, X)Y \right] \mid S=1 \right\} \\
& = E \left\{ c(X)(1-Y)h(a, W, X) \mid S=1 \right\} - \\
& \quad E \left\{ c(X)I(A=a)q(a, Z, X)[(1-Y)h(a, W, X) - Y] \mid S=1 \right\} \\
& \stackrel{Eq(S9)}{=} \theta_{ac} - E \left\{ c(X)I(A=a)q(a, Z, X) \times \right. \\
& \quad \left. E[(1-Y)h(a, W, X) - Y \mid A=a, Z, X, S=1] \mid \right. \\
& \quad \left. S=1 \right\} \\
& \stackrel{Eq(S8)}{=} \theta_{ac} - 0 \\
& = \theta_{ac}.
\end{aligned}$$

A.7 Proof of Theorem 5

To prove Equation (8), we have

$$\begin{aligned}
& E\{(1 - Y)[\kappa_1(A, W, X)q(A, Z, X) - \kappa_1(1, W, X) - \kappa_1(0, W, X)] \mid S = 1\} \\
&= E\{(1 - Y)[\kappa_1(A, W, X)E[q(A, Z, X) \mid A, W, X, Y = 0, S = 1] - \kappa_1(1, W, X) - \\
&\quad \kappa_1(0, W, X)] \mid S = 1\} \\
&\stackrel{Eq(2)}{=} E\{(1 - Y)[\frac{\kappa_1(A, W, X)}{P(A \mid W, X, Y = 0, S = 1)} - \kappa_1(1, W, X) - \kappa_1(0, W, X)] \mid S = 1\} \\
&= E\{(1 - Y)[E\left(\frac{\kappa_1(A, W, X)}{P(A \mid W, X, Y = 0, S = 1)} \mid W, X, Y = 0, S = 1\right) - \kappa_1(1, W, X) - \\
&\quad \kappa_1(0, W, X)] \mid S = 1\} \\
&= E\{(1 - Y)[\kappa_1(1, W, X) + \kappa_1(0, W, X) - \kappa_1(1, W, X) - \kappa_1(0, W, X)] \mid S = 1\} \\
&= 0.
\end{aligned}$$

Equation (9) directly follows Equation (S8).

A.8 Proof of Theorem 6

Under Model 1 and Assumptions 1 and 2, Lemma S1 holds.

- (a) If $q(A, Z, X; \tau)$ is correctly specified and $\hat{\tau} \xrightarrow{p} \tau$, then $q(A, Z, X; \hat{\tau})$ converges to a solution to Equation (2), denoted as $q(A, Z, X)$ with abuse of notation. We have

$$\begin{aligned} \hat{\beta}_{c, \text{PIPW}} &\xrightarrow{p} \log \left(\frac{E \{c(X)I(A=1)Yq(A, Z, X) \mid S=1\}}{E \{c(X)I(A=0)Yq(A, Z, X) \mid S=1\}} \right) \\ &\stackrel{Eq(S6)}{=} \log(\theta_{1c}/\theta_{0c}) \\ &= \beta_0. \end{aligned}$$

- (b) If $h(A, W, X; \psi)$ is correctly specified and $\hat{\psi} \xrightarrow{p} \psi$, then Equation (11) has a unique solution that is ψ_0 such that $h(A, W, X; \psi_0) = h(A, W, X)$, and $h(A, W, X; \hat{\psi})$ is consistent for $h(A, W, X)$. By Law of Large Numbers and Continuous Mapping Theorem, we have

$$\begin{aligned} \hat{\beta}_{c, \text{POR}} &\xrightarrow{p} \log \left(\frac{E \{c(X)(1-Y)h(1, W, X) \mid S=1\}}{E \{c(X)(1-Y)h(0, W, X) \mid S=1\}} \right) \\ &\stackrel{Eq(S9)}{=} \log(\theta_{1c}/\theta_{0c}) \\ &= \beta_0. \end{aligned}$$

- (c) Write $q^*(A, Z, X)$ and $h^*(A, W, X)$ as the probability limit of $q(A, Z, X; \hat{\tau})$ and $h(A, W, X; \hat{\psi})$, respectively. The stated result follows Law of Large Numbers, Continuous Mapping Theorem, and the proof of Theorem 6.

B Estimating conditional causal odds ratio under effect modification by X

In Section 2, we assumed that the effect size of odds ratio is constant across every (U, X) strata. However, in practical situation, the treatment effect may be heterogeneous. Therefore, we relax Model 1 and propose Model S1 below that allows heterogeneous odds ratio across X strata:

Model S1 (Heterogeneous odds ratio model by X).

$$\text{logit } P(Y = 1 \mid A, U, X) = B(X)A + \eta(U, X). \quad (\text{S16})$$

In Model S1, the odds ratio treatment effect $B(X)$ is homogeneous with respect to the latent factors U but is allowed to vary by X . In such scenarios, if the negative control variables are available, the odds ratio function $B(X)$ can still be identified, similar to Theorems 2, 4 and 6, as stated in the following theorem:

Theorem S1. *Under Model S1 and Assumptions 1, 2, 3 and 5, then the odds ratio function $B(X)$ satisfies the following moment equations:*

$$E \left[c(X)I(A = a)Yq(A, Z, X)e^{-B(X)} \mid S = 1 \right] = 0; \quad (\text{S17})$$

$$E \left[c(X)(1 - Y) \left\{ h(1, W, X) - h(0, W, X)e^{B(X)} \right\} \mid S = 1 \right] = 0; \quad (\text{S18})$$

$$E \left\{ c(X) \left[(-1)^{1-A}q(A, Z, X)e^{-B(X)A} \{ Y - (1 - Y)h(A, W, X) \} + (1 - Y) \left\{ h(1, W, X)e^{-B(X)} - h(0, W, X) \right\} \right] \mid S = 1 \right\} = 0. \quad (\text{S19})$$

In cases where a parametric model for the odds ratio function $B(X; \alpha)$ is considered appropriate with a finite dimensional parameter α , similar to the previous PIPW, POR and PDR estimation, the low-dimension parameter α can be estimated by solving the estimating equations

$$\frac{1}{n} \sum_{i=1}^n (-1)^{1-A_i} c(X_i) q(A_i, Z_i, X_i; \tau) Y_i e^{-B(X; \alpha) A_i} = 0; \quad (\text{S20})$$

$$\frac{1}{n} \sum_{i=1}^n c(X_i) (1 - Y_i) \left\{ h(1, W_i, X_i; \hat{\psi}) e^{-B(X; \alpha)} - h(0, W_i, X_i; \hat{\psi}) \right\} = 0; \quad (\text{S21})$$

$$\begin{aligned} \text{or } \frac{1}{n} \sum_{i=1}^n c(X_i) & \left[(-1)^{1-A_i} q(A_i, Z_i, X_i) e^{-B(X; \alpha) A_i} \{Y_i - (1 - Y_i) h(A_i, W_i, X_i)\} + \right. \\ & \left. (1 - Y_i) \left\{ h(1, W_i, X_i) e^{-B(X; \alpha)} - h(0, W_i, X_i) \right\} \right] = 0, \end{aligned} \quad (\text{S22})$$

where $\hat{\tau}$ and $\hat{\psi}$ are estimators for the parameters indexing the treatment and outcome confounding bridge functions, obtained by solving Equations (12) and (13), respectively, and $c(X)$ is a user-specified function with dimension no smaller than that of α .

Alternatively, a similar kernel method to the one in Web Appendix J can be used to obtain a cross-fitting minimax learning estimator of α which solves Equation (S22) and nonparametrically estimates the nuisance functions q and h . With an estimator $\hat{\alpha}$ using either of the above methods, the conditional odds ratio at $X = x$ can then be estimated by $B(x; \hat{\alpha})$.

Nonparametric estimation of $B(X)$ is possible, for example, by using the minimax learning approach (Dikkala et al., 2020) or the semi-nonparametric sieve generalised method of moments approach (Ai and Chen, 2003; Chen, 2007). Such a nonparametric estimator may be of interest, for example, to more flexibly account for the heterogeneity of treatment effects and making flexible prediction on the “individual treatment effect”.

C Regularity conditions for the existence of a solution to Equations (1) and (S7)

The results in this section directly adapted from Appendix B of Cui et al. (2023) and Section B of the Supplementary Materials of Li et al. (2024).

Let $L^2\{F(t)\}$ denote the Hilbert space of all square-integrable functions of t with respect to distribution function $F(t)$, equipped with inner product $\langle g_1, g_2 \rangle = \int g_1(t)g_2(t)dF(t)$. Let $T_{a,u,x} : L^2\{F(z|a, x, Y = 0, S = 1)\} \rightarrow L^2\{F(u|a, x, Y = 0, S = 1)\}$ denote the conditional expectation operator $T_{a,u,x}h = E[h(a, Z, x)|A = a, U = u, X = x, Y = 0, S = 1]$ and let $(\lambda_{a,u,x,n}, \varphi_{a,u,x,n}, \phi_{a,u,x,n})$ denote a singular value decomposition of $T_{a,u,x}$. Similarly, let $T'_{a,u,x} : L^2\{F(w|a, x, Y = 0, S = 1)\} \rightarrow L^2\{F(u|a, x, Y = 0, S = 1)\}$ denote the conditional expectation operator $T'_{a,u,x}g = E[g(a, W, x)|A = a, U = u, X = x, Y = 0, S = 1]$ and let $(\lambda'_{a,u,x,n}, \varphi'_{a,u,x,n}, \phi'_{a,u,x,n})$ denote a singular value decomposition of $T'_{a,u,x}$. Consider the following regularity conditions:

- (1) $\int \int f(z|a, u, x, Y = 0, S = 1)f(u|a, z, x, Y = 0, S = 1)dzdu < \infty$;
- (2) $\int P^{-2}(A = a|U = u, X = x, Y = 0, S = 1)f(u|a, x, Y = 0, S = 1)du < \infty$;
- (3) $\sum_{n=1}^{\infty} \lambda_{a,u,x,n}^{-2} |\langle P^{-1}(A = a|U = u, X = x, Y = 0, S = 1), \phi_{a,u,x,n} \rangle|^2 < \infty$;
- (4) $\sum_{n=1}^{\infty} (\lambda'_{a,u,x,n})^{-2} |\langle P^{-1}(A = a|U = u, X = x, Y = 0, S = 1), \phi'_{a,u,x,n} \rangle|^2 < \infty$.

The solution to Equation (1) exists if Assumption 6(a) and regularity conditions (1), (2) and (3) hold. The solution to Equation (5) exists if Assumption 4(a) and regularity conditions (1), (2) and (4) hold.

D Approximate equivalence of RR and OR for a rare outcome

Under Model 1, the conditional odds ratio is

$$\text{OR}(U, X) = \exp(\beta_0) = \frac{P(Y = 1 \mid A = 1, U, X)P(Y = 0 \mid A = 0, U, X)}{P(Y = 1 \mid A = 0, U, X)P(Y = 1 \mid A = 1, U, X)}.$$

Consider the conditional risk ratio

$$\text{RR}(U, X) := \frac{P(Y = 1 \mid A = 1, U, X)}{P(Y = 1 \mid A = 0, U, X)} = \text{OR}(U, X) \times \frac{P(Y = 0 \mid A = 1, U, X)}{P(Y = 0 \mid A = 0, U, X)}.$$

Under a rare infection assumption that

$$0 \leq P(Y = 1 \mid A = a, U, X) \leq \delta$$

almost surely for $a = 0, 1$ and a small $\delta > 0$, OR and RR satisfies

$$1 - \delta \leq \frac{\text{RR}(U, X)}{\text{OR}(U, X)} = \frac{P(Y = 0 \mid A = 1, U, X)}{P(Y = 0 \mid A = 0, U, X)} \leq \frac{1}{1 - \delta}. \quad (\text{S23})$$

If $\delta \approx 0$, then $\text{OR} \approx \text{RR}$.

Furthermore, denote $Y(a)$ as the potential outcome had the individual received the treatment $A = a$. We make the following standard identifiability assumptions (Hernán and Robins, 2020):

- (a) (Consistency) $Y(a) = Y$ if $A = a$;
- (b) (Confounded exchangeability) $Y(a) \perp\!\!\!\perp A \mid U, X$ for $a = 0, 1$;
- (c) (Positivity) $0 < P(A = 1 \mid U, X) < 1$ almost surely.

Under these assumptions, the marginal causal risk ratio is

$$\begin{aligned}
\text{cRR} &:= \frac{P\{Y(1) = 1\}}{P\{Y(0) = 1\}} \\
&= \frac{\int \int P\{Y(1) = 1 \mid U = u, X = x\} f(u, x) du dx}{\int \int P\{Y(0) = 1 \mid U = u, X = x\} f(u, x) du dx} \\
&\stackrel{(b)}{=} \frac{\int \int P\{Y(1) = 1 \mid A = 1, U = u, X = x\} f(u, x) du dx}{\int \int P\{Y(0) = 1 \mid A = 0, U = u, X = x\} f(u, x) du dx} \\
&\stackrel{(a)}{=} \frac{\int \int P\{Y = 1 \mid A = 1, U = u, X = x\} f(u, x) du dx}{\int \int P\{Y = 1 \mid A = 0, U = u, X = x\} f(u, x) du dx} \\
&= \frac{\int \int \text{RR}(U, X) P\{Y = 1 \mid A = 0, U = u, X = x\} f(u, x) du dx}{\int \int P\{Y = 1 \mid A = 0, U = u, X = x\} f(u, x) du dx} \\
&\stackrel{Eq(S23)}{\leq} \frac{1}{1 - \delta} \exp(\beta_0)
\end{aligned}$$

and

$$\text{cRR} \geq (1 - \delta) \exp(\beta_0).$$

If $\delta \approx 0$, then $\text{cRR} \approx \exp(\beta_0)$.

E Identification of marginal causal risk ratio in a test-negative design with diseased controls

Similar to Schnitzer (2022), we denote D as a categorical variable indicating a subject's outcome, where $D = 2$ indicates $Y = 1$, i.e. the subject is positive for the infection of interest; $D = 1$ indicates that the subject has $Y = 0$ but is positive for another outcome, referred to as the “control infection”, that is known a priori to not be affected by A ; and $D = 0$ indicates that neither outcome is positive. To formalize the assumption of no treatment effect on the control disease, we assume

$$P(D = 1 \mid A = 1, U, X) = P(D = 1 \mid A = 0, U, X). \quad (\text{S24})$$

As sample selection is only restricted to subjects with infection, we have

$$P(D = 0 \mid S = 1, U, X, A) = 0. \quad (\text{S25})$$

Rather than Assumption 1, we make the stronger assumption that

$$A \perp\!\!\!\perp S \mid D, U, X \quad \text{and} \quad A \perp\!\!\!\perp S \mid Y, U, X. \quad (\text{S26})$$

That is, a subject's treatment status has no impact on the sample selection. This assumption holds in a test-negative study if, given a subject's disease status and other traits included in (U, X) , their decision to seek care does not depend on their treatment status.

Under Model 1 and above assumptions, we have

$$\begin{aligned}
\exp(\beta_0) &= \frac{P(A = 1 | Y = 1, U, X)P(A = 0 | Y = 0, U, X)}{P(A = 1 | Y = 0, U, X)P(A = 0 | Y = 1, U, X)} \\
&\stackrel{Eq(S26)}{=} \frac{P(A = 1 | Y = 1, U, X, S = 1)P(A = 0 | Y = 0, U, X, S = 1)}{P(A = 1 | Y = 0, U, X, S = 1)P(A = 0 | Y = 1, U, X, S = 1)} \\
&= \frac{P(A = 1 | D = 2, U, X, S = 1)P(A = 0 | D \neq 2, U, X, S = 1)}{P(A = 1 | D \neq 2, U, X, S = 1)P(A = 0 | D = 2, U, X, S = 1)} \\
&\stackrel{Eq(S25)}{=} \frac{P(A = 1 | D = 2, U, X, S = 1)P(A = 0 | D = 1, U, X, S = 1)}{P(A = 1 | D = 1, U, X, S = 1)P(A = 0 | D = 2, U, X, S = 1)} \\
&\stackrel{Eq(S26)}{=} \frac{P(A = 1 | D = 2, U, X)P(A = 0 | D = 1, U, X)}{P(A = 1 | D = 1, U, X)P(A = 0 | D = 2, U, X)} \\
&= \frac{P(D = 2 | A = 1, U, X)P(D = 1 | A = 0, U, X)}{P(D = 1 | A = 1, U, X)P(D = 2 | A = 0, U, X)} \\
&\stackrel{Eq(S24)}{=} \frac{P(D = 2 | A = 1, U, X)}{P(D = 2 | A = 0, U, X)} \\
&= \frac{P(Y = 1 | A = 1, U, X)}{P(Y = 1 | A = 0, U, X)} \\
&\equiv \text{RR}(U, X)
\end{aligned}$$

Under the standard identifiability assumptions in Section D, the marginal causal risk ratio is

$$\begin{aligned}
\text{cRR} &= \frac{\int \int \text{RR}(U, X) P\{Y = 1 | A = 0, U = u, X = x\} f(u, x) du dx}{\int \int P\{Y = 1 | A = 0, U = u, X = x\} f(u, x) du dx} \\
&= \exp(\beta_0).
\end{aligned}$$

In the above discussion, we do not invoke the rare outcome assumption.

**F Summary of our assumptions and results and those in Li et al.
(2024)**

Table S1: Summary of models, assumptions and identification results, with comparison to those in Li et al. (2024) for identifying conditional risk ratio. All assumption, model, equation, and section numbers are relative to the current manuscript unless otherwise specified.

Results in Li et al. (2024)		
	Risk ratio model approach	PIPW identification
Model	$\log \left(\frac{P(Y = 1 A = 1, U, X)}{P(Y = 1 A = 0, U, X)} \right) = \beta_0$	Model 1
Interpretation of β_0	conditional risk ratio	conditional odds ratio
Selection mechanism	$A \perp\!\!\!\perp S Y, U, X$	Assumption 1
Proximal independence	$Z \perp\!\!\!\perp (Y, S) A, U, X$ $W \perp\!\!\!\perp A U, X$ $W \perp\!\!\!\perp Z A, U, X, Y$ $S \perp\!\!\!\perp Z A, U, X, W, Y$	Assumption 2
q definition	$E\{q(a, Z, X A = a, U, X)\}$ $= 1/P(A = a U, X)$	Equation (1)
h definition	/	/
Completeness	Assumption 4(b)	Assumption 4(b)
Identification	q	Equation (2)
	h	/
	β_0	Equation (3)
Note	(1) Identification of q and β_0 requires a rare outcome condition; (2) Under standard identifiability assumptions in Web Appendix D, β_0 identifies the marginal causal risk ratio. (3) The proximal independence conditions can be replaced by Assumption 2	Under standard identifiability assumptions in Section D, if either the outcome is rare or under the setting of Web Appendix E, β_0 identifies the marginal causal risk ratio.
Results in the current manuscript		
	POR identification	PDR closed-form expression
Model	Model 1	Model 1
Interpretation of β_0	conditional odds ratio	conditional odds ratio
Selection mechanism	Assumption 1	Assumption 1
Proximal independence	Assumption 2	Assumption 2
q definition	/	Equation (1)
h definition	Equation (5)	Equation (5)
Completeness	Assumption 6(b)	Assumptions 4(b), 6(b)
Identification	q	Equation (2) + Assumption 4(b)
	h	Equation (5) + Assumption 6(b)
	β_0	Equation (9)
Note	Under standard identifiability a assumptions in Web Appendix D, if either the outcome is rare or under the setting of Web Appendix E, β_0 identifies the β_0 marginal causal risk ratio.	(1) Under standard identifiability assumptions in Web Appendix D, if either the outcome is rare or under the setting of Web Appendix E, β_0 identifies the marginal causal risk ratio. (2) Equation (9) holds if either q is a solution to Equation (1) or h is a solution to Equation (5).

G Data generating processes with closed-form confounding bridge functions $q(A, Z, X)$ and $h(A, W, X)$

Example 1. If U, Z, W are all binary variable and X is a categorical variable with levels $\{x_1, \dots, x_L\}$, then a suitable model for the confounding bridge functions is the saturated model. That is, the functions $q(a, z, x)$ and $h(a, w, x)$ are completely unrestricted.

Example 2. If Z and W are both multivariate Gaussian random variables conditioning on (A, U, X, Y) , under the setting given in Web Appendix G, suitable models for the confounding bridge functions are

$$q(a, z, x; \tau) = 1 + \exp\{(1 - 2A)(\tau_0 + \tau_A a + \tau_Z^T z + \tau_X^T x)\}; \quad (\text{S27})$$

$$h(a, w, x; \psi) = \exp(\psi_0 + \psi_A a + \psi_W^T w + \psi_X^T x). \quad (\text{S28})$$

where $\tau = (\tau_0, \tau_A, \tau_Z^T, \tau_X^T)^T$ and $\psi = (\psi_0, \psi_A, \psi_Z^T, \psi_X^T)^T$.

Assume that the treatment A and outcome Y both follow a logistic regression model given (U, X) :

$$\begin{aligned} P(A = 1 \mid U, X) &= [1 + \exp\{-(\mu_{0A} + \mu_{UA}^T U + \mu_{XA}^T X)\}]^{-1}, \\ P(Y = 1 \mid A, U, X) &= [1 + \exp\{-(\mu_{0Y} + \beta_0 A + \mu_{UY}^T U + \mu_{XY}^T X)\}]^{-1}. \end{aligned}$$

Further assume the selection indicator S follows a log-linear model of the form:

$$P(S = 1 \mid A, U, X, Y) = \exp(\mu_{AS} A + \mu_{YS} Y + \xi(U, X)).$$

Derivation of $h(A, W, X)$

Assume W is a multivariate Gaussian random variable satisfying

$$W \mid A, U, X, Y = 0 \sim N(\mu_{0W} + \mu_{0UW}^T U + \mu_{0XW}^T X, \Sigma_{0W}).$$

We will show that a suitable model for h is

$$h(a, w, x; \psi) = \exp(\psi_0 + \psi_A a + \psi_W^T w + \psi_X^T x). \quad (\text{S29})$$

We first derive the outcome confounding bridge function h that satisfies Equation (5):

$$E\{h(a, W, X) \mid A = a, U, X, Y = 0, S = 1\} = \\ P(Y = 1 \mid A = a, U, X, S = 1)/P(Y = 0 \mid A = a, U, X, S = 1).$$

We have

$$P(Y = 1 \mid A = a, Z, X)P(S = 1 \mid A = a, Y = 1, Z, X) = \\ \frac{\exp(\mu_{0Y} + \beta_0 a + \mu_{UY}U + \mu_{XY}^T X)}{1 + \exp(\mu_{0Y} + \beta_0 a + \mu_{UY}U + \mu_{XY}^T X)} \exp(\mu_{0S} + \mu_{AS}a + \mu_{YS} + \xi(U, X)) \\ P(Y = 0 \mid A = a, U, X)P(S = 1 \mid A = a, Y = 0, U = 1, U, X) = \\ \frac{1}{1 + \exp(\mu_{0Y} + \beta_0 a + \mu_{UY}U + \mu_{XY}^T X)} \exp(\mu_{0S} + \mu_{AS}a + \xi(U, X)).$$

Therefore

$$P(Y = 1 \mid A = a, U, X, S = 1) \\ = P(Y = 1 \mid A = a, U, X)P(S = 1 \mid A = a, Y = 1, U, X)/\{P(Y = 1 \mid A = a, U, X) \times \\ P(S = 1 \mid A = a, Y = 1, U, X) + P(Y = 0 \mid A = a, U, X)P(S = 1 \mid A = a, Y = 0, U, X)\} \\ = [1 + \exp\{-(\mu_{0Y} + \mu_{YS} + \beta_0 a + \mu_{UY}^T U + \mu_{XY}^T X)\}]^{-1}.$$

and

$$\frac{P(Y = 1 \mid A = a, U, X, S = 1)}{P(Y = 0 \mid A = a, U, X, S = 1)} = \exp(\mu_{0Y} + \mu_{YS} + \beta_0 a + \mu_{UY}^T U + \mu_{XY}^T X).$$

Now assume that $h(a, w, x) = \exp(\psi_0 + \psi_A a + \psi_W w + \psi_X^T x)$. Since

$$W \mid A = a, U, X, Y = 0, S = 1 \sim N(\mu_{0W} + \mu_{0UW}^T U + \mu_{0XW}^T X, \Sigma_{0W}),$$

we have

$$\begin{aligned}
& E\{h(A, W, X) \mid A, U, X, Y = 0, X = 1\} \\
&= \exp(\psi_0 + \psi_A A + \psi_X^T X) E\{\exp(\psi_W^T W) \mid A, U, X, Y = 0, S = 1\} \\
&= \exp(\psi_0 + \psi_A A + \psi_X^T X) \exp(\psi_W^T (\mu_{0W} + \mu_{0UW}^T U + \mu_{0XW}^T X) + \frac{1}{2} \psi_W^T \Sigma_{0W} \psi_W) \\
&= \exp\{(\psi_0 + \psi_W^T \mu_{0W} + \frac{1}{2} \psi_W^T \Sigma_{0W} \psi_W) + \psi_A A + \psi_W^T \mu_{0UW}^T U + (\psi_X^T + \psi_W^T \mu_{0XW}^T) X\}
\end{aligned}$$

Hence Equation (5) holds if we set $\psi = (\psi_0, \psi_A, \psi_W, \psi_X)$ to be

$$\begin{aligned}
\psi_A &= \beta_0, \quad \psi_W = \mu_{0UW}^+ \mu_{UY}, \quad \psi_X = \mu_{XY} - \mu_{0XW} \psi_W = \mu_{XY} - \mu_{UY} \mu_{0UW}^+ \mu_{0XW}, \\
\psi_0 &= \mu_{0Y} + \mu_{YS} - \psi_W^T \mu_{0W} - \frac{1}{2} \psi_W^T \Sigma_{0W} \psi_W \\
&= \mu_{0Y} + \mu_{YS} - \mu_{0UW}^+ \mu_{0W} \mu_{UY} - \frac{1}{2} \mu_{UY}^T (\mu_{0UW}^+)^T \Sigma_{0W} \mu_{0UW}^+ \mu_{UY}.
\end{aligned}$$

Here A^+ denotes the generalised inverse of the matrix A .

Derivation of $q(A, Z, X)$

If Z, W are both multivariate Gaussian random variables satisfying

$$Z \mid A, U, X \sim N(\mu_{0Z} + \mu_{AZ} A + \mu_{UZ}^T U + \mu_{XZ}^T X, \Sigma_Z).$$

While a closed-form parametric model for q does not exist in this example, if the outcome Y is rare in the target population in the sense that

$$P(Y = 1 \mid A = a, U = u, X = x) \approx 0 \quad \text{almost surely,}$$

then the parametric model below may serve as an approximation

$$q(a, z, x; \tau) = 1 + \exp\{(1 - 2A)(\tau_0 + \tau_A a + \tau_Z^T z + \tau_X^T x)\}. \quad (\text{S30})$$

To derive $q(A, Z, X)$, we have

$$\begin{aligned}
& P(A = 1 | U, X)P(Y = 0 | A = 1, U, X)P(S = 1 | A = 1, Y = 0, U, X) \\
&= \frac{\exp(\mu_{0A} + \mu_{UA}^T U + \mu_{XA}^T X)}{1 + \exp(\mu_{0A} + \mu_{UA}^T U + \mu_{XA}^T X)} \frac{1}{1 + \exp(\mu_{0Y} + \beta_0 + \mu_{UY}^T U + \mu_{XY}^T X)} \times \\
&\quad \exp\{\mu_{AS} + \xi(U, X)\}, \\
\text{and} \quad & P(A = 0 | U, X)P(Y = 0 | A = 0, U, X)P(S = 1 | A = 0, Y = 0, U, X) \\
&= \frac{1}{1 + \exp(\mu_{0A} + \mu_{UA}^T U + \mu_{XA}^T X)} \frac{1}{1 + \exp(\mu_{0Y} + \mu_{UY}^T U + \mu_{XY}^T X)} \times \\
&\quad \exp\{\xi(U, X)\}.
\end{aligned}$$

Therefore

$$\begin{aligned}
& P(A = 1 | U, X, Y = 0, S = 1) \\
&= P(A = 1 | U, X)P(Y = 0 | A = 1, U, X)P(S = 1 | Y = 0, S = 1, U, X) / \\
&\quad \{P(A = 1 | U, X)P(Y = 0 | A = 1, U, X)P(S = 1 | Y = 0, S = 1, U, X) + \\
&\quad P(A = 0 | U, X)P(Y = 0 | A = 0, U, X)P(S = 1 | A = 0, Y = 0, U, X)\} \\
&= \frac{\exp(\mu_{0A} + \mu_{AS} + \mu_{UA}^T U + \mu_{XA}^T X)}{\exp(\mu_{0A} + \mu_{AS} + \mu_{UA}^T U + \mu_{XA}^T X) + \frac{1 + \exp(\mu_{0Y} + \beta_0 + \mu_{UY}^T U + \mu_{XY}^T X)}{1 + \exp(\mu_{0Y} + \mu_{UY}^T U + \mu_{XY}^T X)}}
\end{aligned}$$

In the case where $P(Y = 1 | A, U, X)$ are small almost surely, both $\exp(\mu_{0Y} + \beta_0 + \mu_{UY}^T U + \mu_{XY}^T X)$ and $\exp(\mu_{0Y} + \mu_{UY}^T U + \mu_{XY}^T X)$ are necessarily small, then

$$\frac{1 + \exp(\mu_{0Y} + \beta_0 + \mu_{UY}^T U + \mu_{XY}^T X)}{1 + \exp(\mu_{0Y} + \mu_{UY}^T U + \mu_{XY}^T X)} \approx 1,$$

and therefore

$$P(A = 1 | U, X, Y = 0, S = 1) \approx [1 + \exp\{-(\mu_{0A} + \mu_{AS} + \mu_{UA}^T U + \mu_{XA}^T X)\}]^{-1}.$$

We need find a function $q(a, z, x)$ that satisfies Equation (1):

$$E[q(1, Z, X)|A = 1, U, X] = 1 + \exp\{-(\mu_{0A} + \mu_{AS} + \mu_{UA}^T U + \mu_{XA}^T X)\}$$

and

$$E[q(0, Z, X)|A = 0, U, X] = 1 + \exp(\mu_{0A} + \mu_{AS} + \mu_{UA}^T U + \mu_{XA}^T X).$$

We consider $q(A, Z, X) = 1 + \exp\{(1 - 2A)(\tau_0 + \tau_A A + \tau_Z^T Z + \tau_X^T X)\}$, then

$$\begin{aligned} & E[q(1, Z, X)|A = 1, U, X] \\ &= 1 + \exp(-\tau_0 - \tau_A - \tau_X^T X) E[\exp(-\tau_Z^T Z)|A = 1, U, X] \\ &= 1 + \exp(-\tau_0 - \tau_A - \tau_X^T X) \exp\{-\tau_Z^T(\mu_{0Z} + \mu_{AZ} + \mu_{UZ}^T U + \mu_{XZ}^T X) + \frac{1}{2}\tau_Z^T \Sigma_Z \tau_Z\} \\ &= 1 + \exp[-\{(\tau_0 + \tau_A + \tau_Z^T \mu_{0Z} + \tau_Z^T \mu_{AZ} - \frac{1}{2}\tau_Z^T \Sigma_Z \tau_Z) + \tau_Z^T \mu_{UZ}^T U + (\tau_X^T + \tau_Z^T \mu_{XZ}^T)X\}] \end{aligned}$$

and

$$\begin{aligned} & E[q(0, Z, X)|A = 0, U, X] \\ &= 1 + \exp(\tau_0 + \tau_X^T X) E[\exp(\tau_Z^T Z) | A = 0, U, X] \\ &= 1 + \exp(\tau_0 + \tau_X^T X) \exp\{\tau_Z^T(\mu_{0Z} + \mu_{UZ}^T U + \mu_{XZ}^T X) + \frac{1}{2}\tau_Z^T \Sigma_Z \tau_Z\} \\ &= 1 + \exp\{(\tau_0 + \tau_Z^T \mu_{0Z} + \frac{1}{2}\tau_Z^T \Sigma_Z \tau_Z) + \tau_Z^T \mu_{UZ}^T U + (\tau_X^T + \tau_Z^T \mu_{XZ}^T)X\} \end{aligned}$$

Hence τ satisfies

$$\begin{aligned} \tau_Z &= \mu_{UA}^+ \mu_{UZ}, & \tau_X &= \mu_{XA} - \mu_{XZ} \tau_Z = \mu_{XA} - \mu_{XZ} \mu_{UA}^+ \mu_{UZ}, \\ \tau_A &= \tau_Z^T \Sigma_Z \tau_Z, \\ \tau_0 &= \mu_{0A} + \mu_{AS} - \tau_Z^T \mu_{0Z} - \frac{1}{2} \tau_Z^T \Sigma_Z \tau_Z. \end{aligned}$$

Finally, under a strong assumption that the distribution of U follows a location-shift model, we show in Web Appendix H that identification of β_0 can be achieved through a two-stage regression method similar to

that in Liu et al. (2025) where the conditional distributions of W and Y given (A, Z, X) are sequentially estimated through generalized linear regression models, which provides a more computationally efficient algorithm albeit under more restrictive settings.

H Estimation using a two-stage regression approach

To motivate the two-stage regression approach, We assume a location-shift model for the distribution of U .

Assumption S1 (Location-shift model for U).

$$U = E(U \mid A, Z, X, Y = 0, S = 1) + \epsilon$$

where $E(\epsilon) = 0$ and $\epsilon \perp\!\!\!\perp (A, Z, X) \mid Y = 0, S = 1$.

This assumption is similar to Equation (18) in Liu et al. 2023.

Write $\gamma(A, Z, X) = E(U \mid A, Z, X, Y = 0, S = 1)$. By Theorem 3(a), we have

$$\begin{aligned} & \frac{P(Y = 1 \mid A, Z, X, S = 1)}{P(Y = 0 \mid A, Z, X, S = 1)} \\ &= E \{h(A, W, X) \mid A, Z, X, Y = 0, S = 1\} \\ &= E[E \{h(A, W, X) \mid A, U, Z, X, Y = 0, S = 1\} \mid A, Z, X, Y = 0, S = 1] \\ &\stackrel{A.2(b)}{=} E[E \{h(A, W, X) \mid A, U, X, Y = 0, S = 1\} \mid A, Z, X, Y = 0, S = 1] \\ &= E \{ \exp(\mu_{0Y} + \mu_{YS} + \beta_0 A + \mu_{UY} U + \mu_{XY}^T X) \mid A, Z, X, Y = 0, S = 1 \} \\ &= \exp(\mu_{0Y} + \mu_{YS} + \beta_0 A + \mu_{UY}^T \gamma(A, Z, X) + \mu_{XY}^T X) E \{ \exp(\mu_{UY} \epsilon \mid Y = 0, S = 1) \} \\ &= \exp(\mu_{0Y}^* + \beta_0 A + \mu_{UY}^T \gamma(A, Z, X) + \mu_{XY}^T X) \end{aligned} \tag{S31}$$

where

$$\mu_{0Y}^* = \mu_{0Y} + \mu_{YS} + \log E \{ \exp(\mu_{UY} \epsilon \mid Y = 0, S = 1) \}.$$

In the special case of a linear model where $\gamma(A, Z, X) = \gamma_0 + \gamma_A A + \gamma_Z^T Z + \gamma_X^T X$, Equation (S31) reduces to a logistic linear regression model.

We further assume $W \mid A, U, X, Y = 0, S = 1$ follows a linear mean model:

$$E(W \mid A, U, X, Y = 0, S = 1) = c_{01} + c_{U1}^T U + c_{X1}^T X,$$

then by simple algebra and the fact that $W \perp\!\!\!\perp Z \mid A, U, X, Y, S$, we have

$$\begin{aligned}
E(W \mid A, Z, X, Y = 0, S = 1) &= E\{E(W \mid A, U, Z, X, Y = 0, S = 1) \mid A, Z, X, Y = 0, S = 1\} \\
&= E\{E(W \mid A, U, X, Y = 0, S = 1) \mid A, Z, X, Y = 0, S = 1\} \\
&= c_{01} + c_{U1}^T \gamma(A, Z, X) + c_{X1}^T X.
\end{aligned} \tag{S32}$$

When $\gamma(A, Z, X) = \gamma_0 + \gamma_A A + \gamma_Z^T Z + \gamma_X^T X$, Equation (S32) reduces to a linear regression model. Equations (S31) and (S32) jointly suggest a two-stage regression approach to estimate β_0 :

Algorithm 1 Two-Stage Regression with a linear mean model for the negative control outcome

- 1: Fit the linear regression model with W as the outcome and A , Z and X as the covariates using the observations with $Y = 0$ in the selected sample ($S = 1$);
 - 2: Obtain the linear predictor $\hat{E}(W \mid A, Z, X, Y = 0, S = 1)$;
 - 3: Fit the logistic regression model with Y as the outcome and A , $\hat{E}(W \mid A, Z, X, Y = 0, S = 1)$ and X as the covariates.
-

I An alternative set of completeness assumptions to identify β_0

Instead of Assumptions 4(b) and 6(b), we consider the following completeness conditions:

Assumption S2 (Completeness).

- (a) For any squared-integrable function g , if $E[g(U) \mid A, W, X, Y = 0, S = 1]$, then $g(U) = 0$ almost surely;
- (b) For any squared-integrable function g , if $E[g(U) \mid A, Z, X, S = 1]$, then $g(U) = 0$ almost surely.

Notably, Assumption S2(a) is identical to Assumption 6(a).

We claim that under Assumption S2, although Equations (2) and (4.2) may have multiple solutions, each solution is a treatment or outcome confounding bridge function. We formally state this result in the theorem below.

Theorem S2 (Identification of confounding bridge function).

- (a) Under Assumption S2(a), if q is a solution to Equation (2), then q is also a solution to Equation (1);
- (b) Under Assumption S2(b), if h is a solution to Equation (5), then h is also a solution to Equation (4).

Proof. (a) If q solves Equation (2), then for $a = 0, 1$, we have

$$\begin{aligned}
 0 &= E \left\{ q(a, Z, X) - \frac{1}{P(A = a \mid W, X, Y = 0, S = 1)} \mid A = a, W, X, Y = 0, S = 1 \right\} \\
 &= E \left[E \left\{ q(a, Z, X) - \frac{1}{P(A = a \mid W, X, Y = 0, S = 1)} \mid A = a, U, W, X, Y = 0, \right. \right. \\
 &\quad \left. \left. S = 1 \right\} \mid A, W, X, Y = 0, S = 1 \right] \\
 &\stackrel{A.2}{=} E \left[E \left\{ q(a, Z, X) - \frac{1}{P(A = a \mid W, X, Y = 0, S = 1)} \mid A = a, U, X, Y = 0, \right. \right. \\
 &\quad \left. \left. S = 1 \right\} \mid A, W, X, Y = 0, S = 1 \right].
 \end{aligned}$$

By Assumption S2(a), we have

$$E \left\{ q(a, Z, X) - \frac{1}{P(A = a \mid W, X, Y = 0, S = 1)} \mid A = a, U, X, Y = 0, S = 1 \right\} = 0,$$

or equivalently,

$$\begin{aligned}
& E\{q(a, Z, X) \mid A = a, U, X, Y = 0, S = 1\} \\
&= E\left\{\frac{1}{P(A = a \mid W, X, Y = 0, S = 1)} \mid A = a, U, X, Y = 0, S = 1\right\} \\
&= \int \frac{f(w \mid A = a, U, X, Y = 0, S = 1)}{P(A = a \mid W, X, Y = 0, S = 1)} dw \\
&= \int \frac{P(A = a \mid W, X, Y = 0, S = 1)f(w \mid U, X, Y = 0, S = 1)}{P(A = a \mid W, X, Y = 0, S = 1)P(A = a \mid U, X, Y = 0, S = 1)} dw \\
&= \frac{1}{P(A = a \mid U, X, Y = 0, S = 1)}.
\end{aligned}$$

(b) If h solves Equation (5), or equivalently, Equation (S8), then

$$\begin{aligned}
0 &= E\{(1 - Y)h(A, W, X) - Y \mid A, Z, X, S = 1\} \\
&= E[E\{(1 - Y)h(A, W, X) - Y \mid A, U, Z, X, S = 1\} \mid A, Z, X, S = 1] \\
&\stackrel{A.2}{=} E[E\{(1 - Y)h(A, W, X) - Y \mid A, U, X, S = 1\} \mid A, Z, X, S = 1]
\end{aligned}$$

By Assumption S2(b), we have

$$E\{(1 - Y)h(A, W, X) - Y \mid A, U, X, S = 1\} = 0.$$

□

J Cross-fitting kernel learning estimator for nonparametric estimation of β_0 .

Under Model 1 and Assumptions 1 and 2, by Lemma S1, estimation of β_0 reduces to estimation of θ_{ac} . We first derive an influence function of θ_{ac} .

Theorem S3. *An influence function for ζ_{ac} is*

$$IF_{\zeta_{ac}}(O; q, h) = \frac{S}{P(S=1)}(1-Y)c(X)h(a, W, X) - \frac{S}{P(S=1)}I(A=a)q(a, Z, X)c(X)[(1-Y)h(A, W, X) - Y] - \zeta_{ac},$$

Proof. We let $\mu_{ac} = E[Sc(X)I(A=a)Yq(A, Z, X) \mid S=1]$. Because $\mu_{ac} = \zeta_{ac}P(S=1)$, their influence functions $IF_{\mu_{ac}}$ and $IF_{\zeta_{ac}}$ satisfy

$$IF_{\theta_{ac}}(O; q, h) = IF_{\mu_{ac}}(O; q, h)/P(S=1).$$

We proceed to derive $IF_{\mu_{ac}}$.

Consider regular one-dimensional parametric submodel indexed by t , where $t=0$ indicates the truth, we have

$$\frac{\partial \mu_{1ct}}{\partial t} = E[c(X)\dot{q}(A, Z, X)AYS] + E[c(X)q(A, Z, X)AYS R(O)],$$

where R denotes the score function.

To calculate $E[c(X)\dot{q}(A, Z, X)AYS]$, in Equation (S12), taking $n(A, W, X) = Ah(A, W, X)c(X)$, we have

$$\begin{aligned} & E[S(1-Y)c(X)\dot{q}(A, Z, X)Ah(A, W, X)] \\ &= E\{ASc(X)\dot{q}(A, Z, X)E[(1-Y)h(A, W, X) \mid A, Z, X, S]\} \\ &\stackrel{(12)}{=} E\{ASc(X)\dot{q}(A, Z, X)E[Y \mid A, Z, X, S=1]\} \\ &= E[c(X)\dot{q}(A, Z, X)AYS] \end{aligned}$$

In Equation (S14), by taking $n(A, W, X) = Ah(A, W, X)c(X)$, we have

$$\begin{aligned}
& E \left[S(1-Y)c(X) \frac{R(A | W, X, Y=0, S=1)}{f(A | W, X, Y=0, S=1)} Ah(A, W, X) \right] \\
&= E \left[S(1-Y)c(X) \frac{R(O)}{f(A | W, X, Y=0, S=1)} Ah(A, W, X) \right] \\
&\quad - E \left\{ S(1-Y)c(X) R(W, X, Y, S) \frac{Ah(A, W, X)}{f(A | W, X, Y=0, S=1)} \right\} \\
&= E \left[S(1-Y)c(X) \frac{R(O)}{f(A | W, X, Y=0, S=1)} Ah(A, W, X) \right] \\
&\quad - E \left\{ S(1-Y)c(X) R(W, X, Y, S) E \left[\frac{Ah(A, W, X)}{f(A | W, X, Y=0, S=1)} \mid W, X, \right. \right. \\
&\quad \left. \left. Y=0, S=1 \right] \right\} \\
&= E \left[S(1-Y)c(X) \frac{R(O)}{f(A | W, X, Y=0, S=1)} Ah(A, W, X) \right] - \\
&\quad E \{ S(1-Y)c(X) R(W, X, Y, S) h(1, W, X) \} \\
&= E \left\{ S(1-Y)c(X) \left[\frac{Ah(A, W, X)}{f(A | W, X, Y=0, S=1)} - h(1, W, X) \right] R(O) \right\}
\end{aligned}$$

We therefore have

$$\begin{aligned}
& E[c(X)\dot{q}(A, Z, X)AYS] \\
&= E \left\{ S(1-Y)c(X) \left[\frac{1}{f(A | W, X, Y=0, S=1)} - q(A, Z, X) \right] Ah(A, W, X) R(O) \right\} \\
&\quad - E \left\{ S(1-Y)c(X) \left[\frac{Ah(A, W, X)}{f(A | W, X, Y=0, S=1)} - h(1, W, X) \right] R(O) \right\} \\
&= E \{ S(1-Y)c(X) [h(1, W, X) - Aq(A, Z, X)h(A, W, X)] R(O) \}
\end{aligned}$$

Hence

$$\begin{aligned}
\frac{\partial \mu_{1ct}}{\partial t} &= E \left\{ Sc(X) \left((1-Y) \left[h(1, W, X) - Aq(A, Z, X)h(A, W, X) \right] + \right. \right. \\
&\quad \left. \left. Y Aq(A, Z, X) \right) R(O) \right\} \\
&= E \left\{ Sc(X) \left((1-Y) [h(1, W, X) - Aq(A, Z, X)h(A, W, X)] + Y Aq(A, Z, X) - \right. \right. \\
&\quad \left. \left. \mu_{1c} \right) R(O) \right\}
\end{aligned}$$

and

$$IF_{\mu_{1c}}(\cdot) = S(1-Y)c(X)h(1, W, X) - SAq(A, Z, X)c(X) [(1-Y)h(A, W, X) - Y] - \mu_{1c}.$$

Similarly,

$$\begin{aligned}
IF_{\mu_{0c}}(\cdot) &= Sc(X) \left\{ (1-Y) \left[h(0, W, X) - (1-A)q(A, Z, X)h(A, W, X) \right] + \right. \\
&\quad \left. Y(1-A)q(A, Z, X) \right\} \\
&= S(1-Y)c(X)h(0, W, X) - S(1-A)q(A, Z, X)c(X) \left[(1-Y)h(A, W, X) - \right. \\
&\quad \left. Y \right] - \mu_{0c}.
\end{aligned}$$

□

The influence function $IF_{\zeta_{ac}}(O; q, h)$ can be written as

$$\begin{aligned}
IF_{\zeta_{ac}}(O; q_a, h_a) &= \frac{S}{P(S=1)} \{ g_{a1}(A, Y, X)q_a(Z, X)h_a(W, X) + \\
&\quad g_{a2}(Y, X)q_a(Z, X) + g_{a3}(Y, X)h_a(W, X) \} - \zeta_{ac}, \tag{S33}
\end{aligned}$$

where

$$\begin{aligned}
g_{a1}(A, Y, X) &= -c(X)I(A = a)(1 - Y), & g_{a2}(A, Y, X) &= c(X)I(A = a)Y, \\
g_{a3}(A, Y, X) &= c(X)(1 - Y), & h_a(W, X) &= h(a, W, X), \\
\text{and } q_a(Z, X) &= q(a, Z, X).
\end{aligned}$$

This influence function belongs to the class of doubly-robust influence functions considered in Ghassami et al. (2022), and thus their method directly applies. By Theorem 1 of Rotnitzky et al. (2021), such an influence function also satisfies the so-call "mixed-bias" structure, and therefore $\sqrt{n}$ -consistent doubly robust estimation is possible even when the nuisance functions are estimated nonparametrically. In this section, we describe a cross-fitting minimax kernel learning estimator for ζ_{ac} and stated the theoretic property of the resulting estimator of β_0 .

We randomly partition the data evenly into L subsamples. For each $l \in \{1, \dots, L\}$, let I_l and n_l be the indices and size of data in the l th sample, respectively. Let $m_l = n - n_l$ be the size of the sample excluding I_l . Write $q_a(Z, X) = q(a, Z, X)$ and $h_a(W, X) = h(A, W, X)$. As proposed in Ghassami et al. (2022), we estimate the nuisance functions q_a and h_a using data from I_l^c by solving the following optimization problems:

$$\begin{aligned}
\tilde{q}_{al} = \arg \min_{q_a \in \mathcal{Q}_a} \max_{h \in \mathcal{H}_a^*} \frac{1}{m_l} \sum_{i \in I_l^c} & \left[\dot{h}(W_i, X_i) \{q_a(Z_i, X_i)g_{a1}(A_i, Y_i, X_i) + g_3(Y_i, X_i)\} \right. \\
& \left. - \dot{h}^2(W_i, X_i) \right] - \lambda_{\mathcal{H}_a^*}^{q_a} \|\dot{h}\|_{\mathcal{H}_a^*}^2 + \lambda_{\mathcal{Q}_a}^{q_a} \|q_a\|_{\mathcal{Q}_a}^2; \tag{S34}
\end{aligned}$$

$$\begin{aligned}
\tilde{h}_{al} = \arg \min_{h_a \in \mathcal{H}_a} \max_{\dot{q} \in \mathcal{Q}_a^*} \frac{1}{m_l} \sum_{i \in I_l^c} & \left[\dot{q}(Z_i, X_i) \{h_a(W_i, X_i)g_{a1}(A_i, Y_i, X_i) + g_2(A_i, Y_i, X_i)\} \right. \\
& \left. - \dot{q}^2(Z_i, X_i) \right] - \lambda_{\mathcal{Q}_a^*}^{h_a} \|\dot{q}\|_{\mathcal{Q}_a^*}^2 + \lambda_{\mathcal{H}_a}^{h_a} \|h_a\|_{\mathcal{H}_a}^2. \tag{S35}
\end{aligned}$$

where $\mathcal{Q}_a$, $\mathcal{H}_a$, $\mathcal{Q}_a^*$ and $\mathcal{H}_a^*$ are reproducing kernel Hilbert spaces (RKHS) generated by the kernel functions $K_{\mathcal{Q}_a}$, $K_{\mathcal{H}_a}$, $K_{\mathcal{Q}_a^*}$ and $K_{\mathcal{H}_a^*}$, equipped with norms $\|\cdot\|_{\mathcal{Q}_a}$, $\|\cdot\|_{\mathcal{H}_a}$, $\|\cdot\|_{\mathcal{Q}_a^*}$ and $\|\cdot\|_{\mathcal{H}_a^*}$, respectively, and $\lambda_{\mathcal{Q}_a}^{q_a}$, $\lambda_{\mathcal{H}_a}^{h_a}$, $\lambda_{\mathcal{Q}_a^*}^{q_a}$ and $\lambda_{\mathcal{H}_a^*}^{h_a}$ are non-negative regularizing parameters. The above optimization problems have closed form solutions, which we leave to the end of this section.

The objective function for $\tilde{q}_{al}$ can be viewed as a regularised version of empirical perturbation evaluated

using data from I_l^c , by moving the influence function from the pair (q_a, h_a) to $(q_a, h_a + \dot{h})$, where the perturbation is defined as

$$\begin{aligned} prt_{\theta_{ac}}(q_a, h_a; 0, \dot{h}) &= IF_{\theta_{ac}}(A, Y, Z, W, X; q_a, h_a + \dot{h}) - IF_{\theta_{ac}}(A, Y, Z, W, X; q_a, h_a) \\ &= \frac{S}{P(S=1)} \dot{h}(W, X) \{q_a(Z, X)g_{a1}(A, Y, X) - g_{a3}(Y, X)\}. \end{aligned}$$

The objective function contains two forms of regularization: the term $\dot{h}^2$ improves the robustness against model misspecification, stability and the convergence rates, while the Tikhonov-type penalties $\lambda_{\mathcal{H}_a}^{q_a} \|\dot{h}\|_{\mathcal{H}_a}^2$ and $\lambda_{\mathcal{Q}_a}^{q_a} \|q_a\|_{\mathcal{Q}_a}^2$ penalise the complexity of the classes of nuisance functions considered. The objective function for $\tilde{h}_{al}$ can be interpreted similarly.

We estimate ζ_{ac} with the l th subsample with the PDR estimator

$$\begin{aligned} \tilde{\zeta}_{ac,l} &= \frac{1}{n_l} \sum_{i \in I_l} c(X_i) [(1 - Y_i) \tilde{h}_{al}(W_i, X_i) - \\ &\quad I(A_i = a) \tilde{q}_{al}(Z_i, X_i) \{ (1 - Y_i) \tilde{h}_{al}(W_i, X_i) - Y_i \}]. \end{aligned} \quad (\text{S36})$$

The final estimator for ζ_{ac} is $\tilde{\zeta}_{ac} = \sum_{l=1}^L \tilde{\zeta}_{ac,l} / L$, which gives the cross-fitting minimax kernel learning estimator for β_0 as $\tilde{\beta}_0 = \log(\tilde{\zeta}_{1c} / \tilde{\zeta}_{0c})$. We summarise the estimation of ζ_{ac} in Algorithm 2.

Algorithm 2 A cross-fitting minimax kernel learning algorithm to estimate β_0 .

- 1: Split the data into L even subsamples
 - 2: For $a = 0$ to $a = 1$
 - 3: For $l = 1$ to $l = L$
 - 4: Solve $\tilde{q}_{al}$ and $\tilde{h}_{al}$ in Equations (S34) and (S35)
 - 5: Obtain $\tilde{\zeta}_{ac,l}$ according to Equation (S36)
 - 6: $\tilde{\zeta}_{ac} \leftarrow \sum_{l=1}^L \tilde{\zeta}_{ac,l} / L$
 - 7: Output $\tilde{\beta}_0 = \log(\tilde{\zeta}_{1c} / \tilde{\zeta}_{0c})$.
-

The estimator $\tilde{\beta}_{0c}$ is consistent and asymptotically normal, thus allowing valid statistical inference, as stated in Theorem S4 below.

Theorem S4. *Under Model 1 and Assumptions 1-6 and suitable regularity conditions listed in Web Appendix K, the cross-fitting estimator $\tilde{\beta}_0$ in Algorithm 2 is consistent and asymptotically normal of β_0 .*

We prove Theorem S4 in Web Appendix L.

Implementation of this approach is challenging, as the performance of the method proves to be sensitive to the value of tuning parameters. We leave an efficient implementation of this approach to future research.

The closed form solutions for $\tilde{q}_{al}$ and $\tilde{h}_{al}$ are

$$\begin{aligned}\tilde{q}_{al}(z, x) &= \sum_{i \in I_l^c} \gamma_{ali} K_{\mathcal{Q}_a}(Z_i, X_i; z, x), \\ \text{and} \quad \tilde{h}_{al}(w, x) &= \sum_{i \in I_l^c} \alpha_{ali} K_{\mathcal{H}_a}(W_i, X_i; w, x),\end{aligned}$$

where

$$\begin{aligned}\alpha_{al} &= -(K_{\mathcal{H}_a, l} D_{al} \Gamma_{al} D_{al} K_{\mathcal{H}_a, l} + m_l^2 \lambda_{\mathcal{H}_a}^{h_a} K_{\mathcal{H}_a, l})^+ K_{\mathcal{H}_a, l} D_{al} \Gamma_{al} g_{a2, l}, \\ \gamma_{al} &= -(K_{\mathcal{Q}_a, l} D_{al} \Omega_{al} D_{al} K_{\mathcal{Q}_a, l} + m_l^2 \lambda_{\mathcal{Q}_a}^{q_a} K_{\mathcal{Q}_a, l})^+ K_{\mathcal{Q}_a, l} D_{al} \Gamma_{al} g_{a3, l}, \\ \Gamma_{al} &= \frac{1}{4} K_{\mathcal{Q}_a^*, l} \left(\frac{1}{m_l} K_{\mathcal{Q}_a^*, l} + \lambda_{\mathcal{Q}_a^*}^{h_a} I_{al} \right)^{-1}, \\ \Omega_{al} &= \frac{1}{4} K_{\mathcal{H}_a^*, l} \left(\frac{1}{m_l} K_{\mathcal{H}_a^*, l} + \lambda_{\mathcal{H}_a^*}^{q_a} I_{al} \right)^{-1},\end{aligned}$$

D_{al} is a $(n - n_l) \times (n - n_l)$ diagonal matrix with elements $\{g_{a1}(A_i, Y_i, X_i) : i \in I_l^c\}$, $K_{\mathcal{H}, l}$ and $K_{\mathcal{Q}, l}$ are respectively the $(n - n_l) \times (n - n_l)$ empirical kernel matrices with data in the sample I_l^c , $g_{a2, l}$ is the $(n - n_l)$ -vector with elements $\{g_{a2}(Y_i, X_i) : i \in I_l^c\}$, and $g_{a3, l}$ is the $(n - n_l)$ -vector with elements $\{g_{a3}(A_i, Y_i, X_i) : i \in I_l^c\}$.

K Regularity conditions in Theorem S4

For a function class $\mathcal{F}$, let $\mathcal{F}_B := \{f \in \mathcal{F} : \|f\|_{\mathcal{F}}^2 \leq B\}$.

Let $\{\mu_j^{h_a}\}_{j=1}^\infty$ and $\{\varphi_j^{h_a}\}_{j=1}^\infty$ be the eigenvalues and eigenfunctions of the RKHS $\mathcal{H}_a$ and $\{\mu_j^{q_a}\}_{j=1}^\infty$ and $\{\varphi_j^{q_a}\}_{j=1}^\infty$ be the eigenvalues and eigenfunctions of the RKHS $\mathcal{Q}_a$. For any $m \in \mathbb{N}_+$, let $V_m^{\mathcal{H}_a}$ and $V_m^{\mathcal{Q}_a}$ be the $m \times m$ matrices with entry (i, j) defined respectively as

$$[V_m^{\mathcal{H}_a}]_{i,j} = E[E[\varphi_i^{\mathcal{H}_a}(W, X) | Z, X, S = 1]E[\varphi_j^{\mathcal{H}_a}(W, X) | Z, X, S = 1]]$$

and

$$[V_m^{\mathcal{Q}_a}]_{i,j} = E[E[\varphi_i^{\mathcal{Q}_a}(Z, X) | W, X, S = 1]E[\varphi_j^{\mathcal{Q}_a}(Z, X) | W, X, S = 1]].$$

Let $\lambda_m^{\mathcal{H}_a}$ and $\lambda_m^{\mathcal{Q}_a}$ be the minimum eigenvalues of $V_m^{\mathcal{H}_a}$ and $V_m^{\mathcal{Q}_a}$, respectively.

R.1 The functions $g_{a1}(A, Y, X) = c(X)I(A = a)(1 - Y)$, $g_{a2}(A, Y, X) = c(X)I(A = a)Y$ and $g_{a3}(Y, X) = c(X)(1 - Y)$ are bounded.

R.2 There exists a constant σ_1 such that

$$|E[c(X)I(A = a)(1 - Y) | Z, W, X, S = 1]| > \sigma_1 > 0$$

almost surely for $a = 0, 1$.

R.3 The nuisance functions $q(A, Z, X)$ and $h(A, W, X)$ have finite second moments.

R.4 The nuisance functions $q_a(Z, X)$ and their estimators $\tilde{q}_{al}(Z, X)$ and $\tilde{h}_{al}(W, X)$ in each fold of cross-fitting satisfy

$$\min \left\{ \sup_{z,x} |\tilde{q}_{al}(z, x) + \sup_{w,x} h_a(w, x)|, \sup_{z,x} |q_a(z, x) + \sup_{w,x} \tilde{h}_{al}(w, x)| \right\} < \infty.$$

R.5 $\|\tilde{q}_{al} - q_a\|_2 = o_p(1)$, $\|\tilde{h}_{al} - h_a\|_2 = o_p(1)$;

R.6 There exists a constant C such that $\tilde{h}_a \in \mathcal{H}_{aC}$ and $\tilde{q}_a \in \mathcal{Q}_{aC}$ for $a \in \{0, 1\}$.

R.7

$$|E[E[\varphi_i^{\mathcal{H}_a}(W, X) | Z, X, S = 1]E[\varphi_j^{\mathcal{H}_a}(W, X) | Z, X, S = 1]]| \leq c^{\mathcal{H}_a} \lambda_m^{\mathcal{H}_a}$$

and

$$| E[E[\varphi_i^{\mathcal{Q}_a}(Z, X) \mid W, X, S = 1]E[\varphi_j^{\mathcal{Q}_a}(Z, X) \mid W, X, S = 1]] | \leq c^{\mathcal{Q}_a} \lambda_m^{\mathcal{Q}_a}$$

for some constant $c^{\mathcal{H}_a}$ and $c^{\mathcal{Q}_a}$.

R.8

$$\|E[I(A = a)c(X)(1 - Y)\{\tilde{q}_{at}(Z, X) - q_a(Z, X)\} \mid W, X, S = 1]\|_2 = O(n^{-r_q})$$

and

$$\|E[I(A = a)c(X)(1 - Y)\{\tilde{h}_{at}(W, X) - h_a(W, X)\} \mid Z, X, S = 1]\|_2 = O(n^{-r_h}).$$

R.9 $\mu_m^{\mathcal{H}_a} \sim m^{-s_h}$, $\mu_m^{\mathcal{Q}_a} \sim m^{-s_q}$, $\lambda_m^{\mathcal{H}_a} \sim m^{-t_h}$, and $\lambda_m^{\mathcal{Q}_a} \sim m^{-t_q}$, where the decay rates s_h , s_q , t_h and t_q satisfies

$$\max\{r_h + \frac{s_h r_q}{s_h + t_h}, r_q + \frac{s_q r_h}{s_q + t_q}\} > \frac{1}{2}.$$

L Proof of Theorem S4

Under Model 1 and Assumptions 1, 2 and 4, Theorem 2(b) states that $\beta_0 = \log(\zeta_{1c}/\zeta_{0c})$. By delta-method, it suffices to prove that $\tilde{\zeta}_{ac}$ is asymptotically linear with the influence function $IF_{\zeta_{ac}}(O)$, the proof of which follows Ghassami et al. (2022) Theorem 2 and Lemma 1, which we reproduced below.

For $a = 0, 1$ and a given value $\delta > 0$, we define $\mathcal{H}_{aC}^{|\delta|} := \{h_a \in \mathcal{H}_{aC} : \|E[h_a(W, X) \mid Z, X, S = 1]\|_2 \leq \delta\}$ and $\mathcal{Q}_{aC}^{|\delta|} := \{q_a \in \mathcal{Q}_{aC} : \|E[q_a(Z, X) \mid W, X, S = 1]\|_2 \leq \delta\}$. We define the measures of ill-posedness $\tau_{h_a}(\delta) := \sup_{h_a \in \mathcal{H}_{aC}^{|\delta|}} \|h_a\|_2$ and $\tau_{q_a}(\delta) := \sup_{q_a \in \mathcal{Q}_{aC}^{|\delta|}} \|q_a\|_2$.

Lemma S2. (Ghassami et al. (2022) Lemma 1) Under regularity conditions R.6 and R.7, for any $\epsilon > 0$ we have

$$\tau_{h_a}(\epsilon)^2 \leq \min_{m \in \mathbb{N}_+} \left\{ \frac{4\epsilon^2}{\lambda_m^{\mathcal{H}_a}} + [4(c^{\mathcal{H}_a})^2 + 1]C\mu_{m+1}^{\mathcal{H}_a} \right\}$$

and

$$\tau_{q_a}(\epsilon)^2 \leq \min_{m \in \mathbb{N}_+} \left\{ \frac{4\epsilon^2}{\lambda_m^{\mathcal{Q}_a}} + [4(c^{\mathcal{Q}_a})^2 + 1]C\mu_{m+1}^{\mathcal{Q}_a} \right\}.$$

By Lemma S2 and R.9, we have that

$$\tau_{h_a}(\epsilon) = O\left(\frac{s_h}{\epsilon^{s_h} + t_h}\right) \quad \text{and} \quad \tau_{q_a}(\epsilon) = O\left(\frac{s_q}{\epsilon^{s_q} + t_q}\right)$$

The condition R.9 further implies that

$$\begin{aligned} \min\{n^{-r_h} \tau_{h_a}(n^{-r_q}), n^{-r_q} \tau_{q_a}(n)\} &= O\left(\min\left\{n^{-\left(r_h + \frac{s_h r_q}{s_h + t_h}\right)}, n^{-\left(r_q + \frac{s_q r_h}{s_q + t_q}\right)}\right\}\right) \\ &= o(n^{-1/2}). \end{aligned} \tag{S37}$$

The result follows by the theorem below:

Theorem S5. (Ghassami et al. (2022) Theorem 2) Under regularity conditions R.1-R.6 and R.8, if Equation (S37) holds in addition, then the estimator $\tilde{\theta}_{ac}$ is asymptotically linear with the influence function $IF_{\zeta_{ac}}(O)$.

M Extension to polytomous treatment and effect

Suppose we want to study the effect of a polytomous treatment A on a polytomous outcome Y , where A have levels $a_0, a_1, \dots, a_J$ and Y have levels $y_0, \dots, y_K$. Here a_0 and y_0 denote the reference treatment and outcome, respectively. We make the following homogeneity assumption:

Model S2 (Homogeneous odds ratio model).

$$\log \left(\frac{P(Y | A, U, X)}{P(Y = y_0 | A, U, X)} \right) = \sum_{\substack{j \in \{1, \dots, J\} \\ k \in \{1, \dots, K\}}} \beta_j^k I(A = a_j, Y = y_k) + \eta^k(U, X) I(Y = y_k).$$

Model S2 indicates that the odds ratio of $Y = y_k$ relative to $Y = y_0$ against the treatment $A = a_j$ versus $A = a_0$ is β_j^k across all strata of (U, X) . Model S2 is a natural semiparametric extension to the polytomous logistic regression model (Engel, 1988). When $J = 1$ and $K = 1$, Model S2 reduces to Model 1. The goal is to estimate and make inference on every β_j^k in the presence of latent factors U and confounded outcome-dependent sampling.

We make the following assumptions, corresponding to Assumption 1, 3 and 5:

Assumption S3 (No effect modification by A on the outcome-dependent selection). *For $a = a_0, \dots, a_J$ and real-valued unknown functions $\xi^1(U, X), \dots, \xi^K(U, X)$, the sampling mechanism satisfies*

$$\frac{P(S = 1 | Y = y_k, A = a, U, X)}{P(S = 1 | Y = y_0, A = a, U, X)} = \xi^k(U, X)$$

for $k = 1, \dots, K$.

Assumption S4 (Confounding bridge functions). *The exists a treatment confounding bridge function $q(A, Z, X)$ and K outcome confounding bridge functions $h_k(A, W, X)$ such that for $k = 1, \dots, K$ and $a = a_0, a_1, \dots, a_J$,*

$$E\{q(a, Z, X) | A = a, U, X\} = 1/P(A = a | U, X, Y = y_0, S = 1) \quad (\text{S38})$$

$$E\{I(Y = y_0)h_k(a, W, X) | A = a, U, X, S = 1\} = P(Y = y_k | A = a, U, X, S = 1). \quad (\text{S39})$$

We state the following results for inference of β_j^k . The proofs are similar to those in Web Appendix A and are therefore omitted.

Lemma S3. Under Model S2 and Assumption S3, the log odds ratio β_j^k satisfies

$$\beta_j^k = \log(\theta_{jc}^k / \theta_{0c}^k),$$

where

$$\theta_{jc}^k = E \left\{ c(X) \frac{P(A = a_j \mid U, X, Y = y_k, S = 1)}{P(A = a_j \mid U, X, Y = y_0, S = 1)} P(Y = y_k \mid U, X, S = 1) \mid S = 1 \right\}.$$

and $c(X)$ is an arbitrary real-valued function that satisfies $\theta_{jc}^k \neq 0$ for $j = 0, 1, \dots, K$.

Theorem S6 (Identification of $q(A, Z, X)$ and $h_k(A, W, X)$). The functions $q(\cdot)$ and $h_k(\cdot)$, $k = 1, \dots, K$, satisfy the moment conditions

$$E \left[I(Y = y_0) \left\{ \kappa_1^*(A, W, X) q(A, Z, X) - \sum_{j=0}^J \kappa_1^*(a_j, W, X) \right\} \mid S = 1 \right] = 0$$

$$E [\kappa_{2k}^*(A, Z, X) \{ I(Y = y_0) h_k(A, W, X) - I(Y = y_k) \} \mid S = 1] = 0, \quad k=1, \dots, K,$$

where $\kappa_1^*(\cdot)$ and κ_{2k} are arbitrary functions.

Theorem S7 (Identification of θ_{jc}^k). Assume Model S2 and Assumptions S3 and 2 hold, then

(a) if there exists a function q that solves Equation (S38), then

$$\theta_{jc}^k = E \{ I(A = a_j, Y = y_k) c(X) q(a_j, Z, X) \mid S = 1 \};$$

(b) if there exists functions h^k , $k = 1, \dots, K$ that solve Equation (S39), then

$$\theta_{jc}^k = E \{ c(X) I(Y = y_0) h_k(a_j, W, X) \mid S = 1 \};$$

(c) if either (i) the function q solves Equation (S38), or (ii) the functions h_k 's solve Equation (S39), then

$$\theta_{jc}^k = E \{ c(X) [I(Y = y_0) h_k(a_j, W, X) - I(A = a_j) q(a_j, Z, X) \{ I(Y = y_0) h_k(a_j, W, X) - I(Y = y_k) \}] \mid S = 1 \}.$$

By Theorem S6 and S7, to estimate β_j^k one may:

1. Specify suitable parametric working models $q(A, Z, X; \tau)$ and $h_k(A, W, X; \psi_k)$;
2. Estimate the nuisance parameters τ and ψ by solving

$$\begin{aligned} \sum_{i=1}^n I(Y_i = y_0) \left\{ \kappa_1^*(A_i, W_i, X_i) q(A_i, Z_i, X_i; \tau) - \sum_{j=0}^J \kappa_1^*(a_j, W_i, X_i) \right\} &= 0; \\ \sum_{i=1}^n \kappa_{2k}^*(A_i, Z_i, X_i) \{ I(Y_i = y_0) h_k(A_i, W_i, X_i; \psi_k) - I(Y_i = y_k) \} &= 0 \end{aligned}$$

for $k = 1, 2, \dots, K$, where κ_1^* and κ_{2k}^* are user-specified functions that dimensions at least as large as that of τ and ψ_k respectively. Denote the resulting estimator as $\hat{\tau}$ and $\hat{\psi}_k$.

3. Estimate θ_{jc}^k by

$$\begin{aligned} \hat{\theta}_{jc, \text{PIPW}}^k &= \frac{1}{n} \sum_{i=1}^n I(A = a_j, Y = y_k) c(X_i) q(a_j, Z_i, X_i; \hat{\tau}); \\ \hat{\theta}_{jc, \text{POR}}^k &= \frac{1}{n} \sum_{i=1}^n c(X_i) I(Y_i = y_0) h_k(a_j, W_i, X_i; \hat{\psi}_k); \\ \text{or } \hat{\theta}_{jc, \text{PDR}}^k &= \frac{1}{n} \sum_{i=1}^n c(X_i) \left[I(Y_i = y_0) h_k(a_j, W_i, X_i; \hat{\psi}_k) - \right. \\ &\quad \left. I(A_i = a_j) q(a_j, Z_i, X_i; \hat{\tau}) \{ I(Y_i = y_0) h_k(a_j, W_i, X_i; \hat{\psi}_k) - I(Y_i = y_k) \} \right] \end{aligned}$$

for $j = 0, 1, \dots, J$ and $k = 1, \dots, K$ and a user-specified real-valued function $c(X)$ (one may simply set $c(X) = 1$).

4. The resulting estimators for β_j^K include the PIPW estimator $\hat{\beta}_{j, \text{PIPW}}^k = \log(\hat{\theta}_{jc, \text{PIPW}}^k / \hat{\theta}_{0c, \text{PIPW}}^k)$, the POR estimator $\hat{\beta}_{j, \text{POR}}^k = \log(\hat{\theta}_{jc, \text{POR}}^k / \hat{\theta}_{0c, \text{POR}}^k)$, and the PDR estimator $\hat{\beta}_{j, \text{PDR}}^k = \log(\hat{\theta}_{jc, \text{PDR}}^k / \hat{\theta}_{0c, \text{PDR}}^k)$.

Under Model S2 and Assumptions 2 and S3, the PDR estimator satisfies doubly robustness, i.e. $\hat{\beta}_{j, \text{PDR}}^k \xrightarrow{P} \beta_j^k$ if either (i) the parametric working model $q(A, Z, X; \tau)$ is correctly specified, and the completeness assumption 4(b) holds, or (ii) the parametric working model $h(A, W, X; \psi)$ is correctly specified, and Assumption 6(b) holds.

The kernel learning approach in Web Appendix J can similarly be employed to obtain semiparametric estimators for β_j^k with flexible modeling for q and h_k .

N Details of the simulation studies

We generate the data for a target population of N observations according to the following five scenarios

Scenario I: Univariate binary U , Z and W

$$\begin{aligned}U &\sim \text{Bernoulli}(0, 0.5); \\A \mid U &\sim \text{Bernoulli}(\text{expit}(0.2 + 0.4U)) \\Z \mid A, U &\sim \text{Bernoulli}(0.2 + 0.1A + 0.4U + 0.2AU); \\W \mid U &\sim \text{Bernoulli}(0.2 + 0.4U); \\Y \mid A, U &\sim \text{Bernoulli}(\text{expit}(-0.405 - 1.609A - 0.7U)) \\S \mid U, A, Y &\sim \text{Bernoulli}(\exp(-1.7 + 0.2A + 0.4Y + 0.7U)).\end{aligned}$$

Scenario II-V: Continuous U , X , Z , W .

$$\begin{aligned}U &\sim \text{Uniform}(0, 1); \\X &\sim \text{Uniform}(0, 1); \\A \mid U, X &\sim \text{Bernoulli}(\text{expit}(-1 + U + X)) \\Z \mid A, U, X &\sim \text{N}(0.25A + 2X + 4U, 0.25^2); \\W \mid Y, U, X &\sim \text{N}(0.25Y + 2X + 4U, 0.25^2); \\Y \mid A, U, X &\sim \text{Bernoulli}(\text{expit}(-3.892 + \beta_0 A - 2U + 2X)) \\S \mid U, X, A, Y &\sim \text{Bernoulli}(\exp(-5 + 0.4A + 4Y + 0.3U + 0.2X)).\end{aligned}$$

O Detailed simulation results

Additional simulation results in Table S2 include the summary statistics of the oracle estimator (adjusted odds ratio from a logistic regression model of Y on (A, U, X)) and relative efficiencies of the proximal estimators (that is, the standard deviation of the oracle estimator divided by that of the POR, PIPW, or PDR estimator).

Table S2: Bias, standard deviation, relative efficiency, relative efficiency, standard error estimates, and coverage of 95% confidence intervals of the oracle estimator, proximal outcome regression (POR), proximal inverse probability weighted (PIPW), and proximal doubly robust (PDR) estimator for five different scenarios summarized over 10,000 Monte Carlo samples. In Scenario I, the variables (U, Z, W) are univariate binary; in Scenario II-V, the variables (X, U, Z, W) are univariate continuous variables. For each scenario, the median and range of the sample size are reported. Summary statistics of the standard error estimates include the median and 25th, 75th percentiles.

	N	n	Oracle					POR				
			Bias	SD	Rel. Eff.	SE	Coverage	Bias	SD	Rel. Eff.	SE	Coverage
I	3500	1164(1058,1254)	-0.003	0.169	/	0.170(0.166,0.175)	0.958	0.031	0.262	0.646	0.238(0.209,0.295)	0.973
I	5000	1663(1546,1774)	-0.005	0.137	/	0.143(0.140,0.146)	0.958	0.022	0.202	0.677	0.198(0.177,0.232)	0.974
II	1×10^5	1889(1746,2031)	-0.004	0.112	/	0.113(0.111,0.114)	0.951	-0.005	0.124	0.907	0.127(0.124,0.129)	0.957
II	2×10^5	3780(3588,3986)	-0.002	0.079	/	0.080(0.079,0.080)	0.955	-0.003	0.088	0.895	0.089(0.088,0.090)	0.954
III	1×10^5	1889(1746,2031)	-0.003	0.112	/	0.113(0.111,0.114)	0.950	-0.004	0.123	0.907	0.127(0.124,0.129)	0.958
III	2×10^5	3780(3588,3986)	-0.002	0.079	/	0.080(0.079,0.080)	0.953	-0.003	0.089	0.889	0.089(0.088,0.090)	0.950
IV	1×10^5	1889(1746,2031)	-0.004	0.112	/	0.113(0.111,0.114)	0.950	-0.122	0.126	0.887	0.126(0.123,0.131)	0.863
IV	2×10^5	3780(3588,3986)	-0.002	0.079	/	0.080(0.079,0.080)	0.954	-0.121	0.091	0.873	0.090(0.088,0.092)	0.737
V	1×10^5	1889(1746,2031)	-0.003	0.112	/	0.113(0.111,0.114)	0.949	-0.122	0.127	0.885	0.126(0.123,0.131)	0.864
V	2×10^5	3780(3588,3986)	-0.001	0.079	/	0.080(0.079,0.080)	0.955	-0.121	0.091	0.868	0.090(0.088,0.092)	0.738
	N	n	PIPW					PDR				
			Bias	SD	Rel. Eff.	SE	Coverage	Bias	SD	Rel. Eff.	SE	Coverage
I	3500	1164(1058,1254)	0.032	0.262	0.646	0.238(0.209,0.295)	0.973	0.031	0.261	0.646	0.238(0.209,0.295)	0.973
I	5000	1663(1546,1774)	0.022	0.202	0.677	0.198(0.177,0.232)	0.975	0.022	0.202	0.677	0.198(0.177,0.232)	0.975
II	1×10^5	1889(1746,2031)	-0.007	0.118	0.953	0.118(0.116,0.120)	0.947	-0.004	0.119	0.948	0.119(0.117,0.121)	0.950
II	2×10^5	3780(3588,3986)	-0.002	0.082	0.958	0.083(0.082,0.084)	0.956	-0.002	0.082	0.958	0.084(0.083,0.085)	0.954
III	1×10^5	1889(1746,2031)	0.162	0.110	1.020	0.110(0.108,0.112)	0.695	-0.004	0.119	0.941	0.120(0.117,0.122)	0.951
III	2×10^5	3780(3588,3986)	0.165	0.076	1.034	0.078(0.077,0.079)	0.431	-0.003	0.083	0.948	0.084(0.083,0.086)	0.954
IV	1×10^5	1889(1746,2031)	-0.007	0.117	0.953	0.118(0.116,0.120)	0.948	-0.007	0.118	0.944	0.118(0.116,0.121)	0.948
IV	2×10^5	3780(3588,3986)	-0.003	0.083	0.956	0.083(0.082,0.084)	0.953	-0.003	0.083	0.948	0.084(0.082,0.085)	0.953
V	1×10^5	1889(1746,2031)	0.162	0.110	1.019	0.110(0.108,0.112)	0.695	0.177	0.113	0.994	0.112(0.110,0.115)	0.656
V	2×10^5	3780(3588,3986)	0.166	0.076	1.030	0.078(0.077,0.079)	0.425	0.181	0.078	1.001	0.080(0.078,0.081)	0.372

P Discussion of the UMHS data analysis

In our data analysis of Section 9, the proposed PIPW, POR, and PDR methods produced similar VE estimates for Pfizer-BioNTech and Johnson & Johnson vaccines. Similar to Scenario II in the simulation of Section 7, this suggests that the treatment and outcome confounding bridge functions are approximately correctly specified and the proximal estimators are likely unbiased. The Moderna VE estimate using PIPW method is slightly higher than that using POR or PDR method, which could result from model misspecification of the treatment confounding bridge function. The proximal VE estimates are notably higher than those from a logistic regression. They are also closer to the results of previous randomized clinical trials, in which the VE estimates against symptomatic COVID-19 illness were 95.0%, 94.1%, and 66.9% for the Pfizer-BioNTech, Moderna, and J & J vaccines, respectively (Polack et al., 2020; Baden et al., 2021; Sadoff et al., 2021). The remaining differences between our estimates and VE estimates from the clinical trials may result from the fact that the previous clinical trials focused on VE against symptomatic illness, while in our analysis, the proximal estimators targeted VE against test-positive SARS-Cov-2 infections.

The UMHS data analysis has several limitations. We did not adjust for previous SARS-CoV-2 infection as it was not available before April 5, 2021 in the UMHS data. However, since the study period was during the early COVID-19 pandemic, it is likely that only a small proportion of the subjects had previous infection and the resulting bias is small. Secondly, as is typical with EHR-based studies, the UMHS data did not record failed SARS-CoV-2 test and might not capture all subjects' vaccination history, which may lead to underestimation of both the infection and vaccination rates. We expect both errors to be rare and not cause substantial bias to our analysis.

Q Interpretation of β_0 without homogeneity assumptions

For generality, we consider the interpretation of $\beta_0(X)$ under Model S1, defined as a solution to Equation (S17) in Theorem S1 in Web Appendix B, which has a closed-form expression

$$\begin{aligned}\beta_0(X) &= \log \left(\frac{E\{AYq(A, Z, X) \mid X, S = 1\}}{E\{(1 - A)Yq(A, Z, X) \mid X, S = 1\}} \right) \\ &= \log \left(\frac{E\{SAYq(A, Z, X) \mid X\}}{E\{S(1 - A)Yq(A, Z, X) \mid X\}} \right).\end{aligned}$$

The solution β_0 to Equation (3.3) satisfies

$$\exp(\beta_0) = E[\exp\{\beta_0(X)\}\omega(X)],$$

where

$$\omega(X) = \frac{c(X)E\{S(1 - A)Yq(A, Z, X) \mid X\}}{E\{c(X)S(1 - Y)Aq(A, Z, X)\}}.$$

That is, β_0 can be viewed as a weighted average of $\beta_0(X)$ by weights $\omega(X)$.

Under Model S1 and Assumption 1, we have shown that $\beta_0(x)$ estimates the conditional log odds ratio given $X = x$ in every U stratum of the target population. Such interpretation is invalid if Model S1 does not hold. However, in the two settings discussed below, we shall show that $\beta_0(X)$ still approximates well-known causal estimands.

Q.1 Setting I: rare outcome and treatment-free selection

Instead of Assumption 1, we assume the selection mechanism satisfies the stricter condition that A has no causal effect on sample selection S except through Y . This is the “treatment-free selection” assumption made in Li et al. (2024). This assumption holds in the setting represented in Figure 1(b).

Assumption S5 (Treatment-free selection). $S \perp\!\!\!\perp A \mid Y, U, X$.

We also make the same rare outcome assumption as in Web Appendix D.

Assumption S6 (Rare outcome). $0 < P(Y = 1 \mid A, U, X) < \delta$ almost surely for some small $\delta > 0$.

Writing $S(a)$ as the counterfactual selection indicator were a subject given the treatment status $A = a$, possibly contrary to reality. We make the standard identifiability assumptions as below.

Assumption S7. *The following conditions hold:*

- (a) *(Consistency)* $Y = Y(A)$ and $S = S(A)$ almost surely;
- (b) *(Latent exchangeability)* $(S(a), Y(a)) \perp\!\!\!\perp A$ for $a = 0, 1$.
- (c) *(Positivity)* $0 < P(A = 1 \mid U, X) < 1$ almost surely.

Under Assumptions 2, S5, S6, S7 and the treatment confounding bridge function assumption 3, we can show that $\exp\{\beta_0(x)\} \approx P\{S(1)Y(1) = 1 \mid X = x\}/P\{S(0)Y(0) = 1 \mid X = x\}$, which is the conditional causal risk ratio of being positive for the outcome and selected into the sample. For example, in a test-negative design study of vaccine effectiveness for flu infection (Jackson and Nelson, 2013; Chua et al., 2020), $\beta_0(x)$ approximates the log causal risk ratio of “testing positive for flu infection” in the stratum

$X = x$ of the target population. This is because

$$\begin{aligned}
\beta_0(X) &= \log \left(\frac{E\{SAYq(A, Z, X) \mid X\}}{E\{S(1-A)Yq(A, Z, X) \mid X\}} \right) \\
&= \log \left(\frac{E[E\{SAYq(A, Z, X) \mid U, X\} \mid X]}{E[E\{S(1-A)Yq(A, Z, X) \mid U, X\} \mid X]} \right) \\
&= \log \left(\frac{E\{E[SAYE\{q(A, Z, X) \mid A=1, U, X, Y=0, S=1\} \mid U, X] \mid X\}}{E\{E[S(1-A)YE\{q(A, Z, X) \mid A=0, U, X, Y=1, S=1\} \mid U, X] \mid X\}} \right) \\
&\stackrel{A.2}{=} \log \left(\frac{E\{E[SAYE\{q(A, Z, X) \mid A=1, U, X\} \mid U, X] \mid X\}}{E\{E[S(1-A)YE\{q(A, Z, X) \mid A=0, U, X\} \mid U, X] \mid X\}} \right) \\
&\stackrel{A.3}{=} \log \left(\frac{E[E\{SAY/P(A=1 \mid U, X, Y=0, S=1) \mid U, X\} \mid X]}{E[E\{S(1-A)Y/P(A=0 \mid U, X, Y=0, S=1) \mid U, X\} \mid X]} \right) \\
&= \log \left(\frac{E \left[E \left\{ \frac{P(A=1 \mid U, X)E(SY \mid A=1, U, X)}{P(A=1 \mid U, X, Y=0, S=1)} \mid U, X \right\} \mid X \right]}{E \left[E \left\{ \frac{P(A=0 \mid U, X)E(SY \mid A=0, U, X)}{P(A=0 \mid U, X, Y=0, S=1)} \mid U, X \right\} \mid X \right]} \right) \\
&\stackrel{A.S5}{=} \log \left(\frac{E \left[E \left\{ \frac{P(A=1 \mid U, X)E(SY \mid A=1, U, X)}{P(A=1 \mid U, X, Y=0)} \mid U, X \right\} \mid X \right]}{E \left[E \left\{ \frac{P(A=0 \mid U, X)E(SY \mid A=0, U, X)}{P(A=0 \mid U, X, Y=0)} \mid U, X \right\} \mid X \right]} \right) \\
&= \log \left(\frac{E \left[E \left\{ \frac{P(Y=0 \mid U, X)E(SY \mid A=1, U, X)}{P(Y=0 \mid A=1, U, X)} \mid U, X \right\} \mid X \right]}{E \left[E \left\{ \frac{P(Y=0 \mid U, X)E(SY \mid A=0, U, X)}{P(Y=0 \mid A=0, U, X)} \mid U, X \right\} \mid X \right]} \right) \\
&\stackrel{A.S6}{\leq} \log \left(\frac{E \left[E \left\{ \frac{E(SY \mid A=1, U, X)}{1-\delta} \mid U, X \right\} \mid X \right]}{E[E\{(1-\delta)E(SY \mid A=0, U, X) \mid U, X\} \mid X]} \right) \\
&= \log \left(\frac{E[E\{E(SY \mid A=1, U, X) \mid U, X\} \mid X]}{E[E\{E(SY \mid A=0, U, X) \mid U, X\} \mid X]} \right) - 2\log(1-\delta) \\
&\stackrel{A.S7}{=} \log \left(\frac{E[E\{S(1)Y(1) \mid U, X\} \mid X]}{E[E\{S(0)Y(0) \mid U, X\} \mid X]} \right) - 2\log(1-\delta) \\
&= \log \left(\frac{E\{S(1)Y(1) \mid X\}}{E\{S(0)Y(0) \mid X\}} \right) - 2\log(1-\delta)
\end{aligned}$$

and similarly

$$\beta_0(X) > \log \left(\frac{E\{S(1)Y(1) \mid X\}}{E\{S(0)Y(0) \mid X\}} \right) + 2\log(1-\delta).$$

If $\delta \approx 0$, then

$$\exp\{\beta_0(X)\} \approx \frac{E\{S(1)Y(1) \mid X\}}{E\{S(0)Y(0) \mid X\}} = \frac{P\{S(1)Y(1) = 1 \mid X\}}{P\{S(0)Y(0) = 1 \mid X\}}.$$

Q.2 Setting II: rare outcome and U -independent control sampling

In Setting I, Assumption S5 precludes any treatment-induced sampling. Consider the selection mechanism that satisfies the assumption below instead:

Assumption S8. *The selection mechanism satisfies:*

- (a) $\frac{P(S = 1 \mid A, Y = 1, U, X)}{P(S = 1 \mid A, Y = 0, U, X)} = \xi'(X)$ *almost surely*;
- (b) $P(S = 1 \mid Y = 0, U, X) = \zeta(X)$ *almost surely*.

Assumption S8(a) is stronger than Assumption 2 and requires that neither A nor U modifies the effect of the outcome Y on the selection S on the risk ratio scale, while Assumption S8(b) requires that the probability of control subjects being selected into the sample does not depend on U . This may hold if, for example, in a case-control study where control subjects are manually selected by the analyst only based on measured covariates X . We also

Under Assumptions 2, the rare outcome Assumption S6, S7 and S8, we can show that $\exp\{\beta_0(X)\} \approx$

$P\{Y(1) = 1 \mid X\}/P\{Y(0) = 1 \mid X\}$, the conditional causal risk ratio given X . This is because

$$\begin{aligned}
& \beta_0(X) \\
&= \log \left(\frac{E\{SAYq(A, Z, X) \mid X\}}{E\{S(1-A)Yq(A, Z, X) \mid X\}} \right) \\
&= \log \left(\frac{E[E\{SAYq(A, Z, X) \mid U, X\} \mid X]}{E[E\{S(1-A)Yq(A, Z, X) \mid U, X\} \mid X]} \right) \\
&= \log \left(\frac{E\{E[SAYE\{q(A, Z, X) \mid A=1, U, X, Y=0, S=1\} \mid U, X] \mid X\}}{E\{E[S(1-A)YE\{q(A, Z, X) \mid A=0, U, X, Y=1, S=1\} \mid U, X] \mid X\}} \right) \\
&\stackrel{A.2}{=} \log \left(\frac{E\{E[SAYE\{q(A, Z, X) \mid A=1, U, X\} \mid U, X] \mid X\}}{E\{E[S(1-A)YE\{q(A, Z, X) \mid A=0, U, X\} \mid U, X] \mid X\}} \right) \\
&\stackrel{A.3}{=} \log \left(\frac{E \left\{ \frac{P(A=1 \mid U, X)P(Y=1 \mid A=1, U, X)P(S=1 \mid A=1, Y=1, U, X)}{P(A=1 \mid U, X, Y=0, S=1)} \mid X \right\}}{E \left\{ \frac{P(A=0 \mid U, X)P(Y=1 \mid A=0, U, X)P(S=1 \mid A=0, Y=1, U, X)}{P(A=0 \mid U, X, Y=0, S=1)} \mid X \right\}} \right) \\
&= \log \left(\frac{E \left\{ \frac{P(A=1 \mid U, X)P(Y=1 \mid A=1, U, X)P(S=1 \mid A=1, Y=1, U, X)P(S=1 \mid Y=0, U, X)}{P(A=1 \mid U, X, Y=0)P(S=1 \mid A=1, Y=0, U, X)} \mid X \right\}}{E \left\{ \frac{P(A=0 \mid U, X)P(Y=1 \mid A=0, U, X)P(S=1 \mid A=0, Y=1, U, X)P(S=1 \mid Y=0, U, X)}{P(A=0 \mid U, X, Y=0)P(S=1 \mid A=0, Y=0, U, X)} \mid X \right\}} \right) \\
&\stackrel{A.S8}{=} \log \left(\frac{E \left\{ \frac{P(A=1 \mid U, X)P(Y=1 \mid A=1, U, X)}{P(A=1 \mid U, X, Y=0)} \mid X \right\} \xi'(X)\zeta(X)}{E \left\{ \frac{P(A=0 \mid U, X)P(Y=1 \mid A=0, U, X)}{P(A=0 \mid U, X, Y=0)} \mid X \right\} \xi'(X)\zeta(X)} \right) \\
&= \log \left(\frac{E \left\{ \frac{P(A=1 \mid U, X)P(Y=1 \mid A=1, U, X)}{P(A=1 \mid U, X, Y=0)} \mid X \right\}}{E \left\{ \frac{P(A=0 \mid U, X)P(Y=1 \mid A=0, U, X)}{P(A=0 \mid U, X, Y=0)} \mid X \right\}} \right) \\
&\stackrel{A.S7}{=} \log \left(\frac{E \left\{ \frac{P(A=1 \mid U, X)P(Y(1)=1 \mid U, X)}{P(A=1 \mid U, X, Y=0)} \mid X \right\}}{E \left\{ \frac{P(A=0 \mid U, X)P(Y(0)=1 \mid U, X)}{P(A=0 \mid U, X, Y=0)} \mid X \right\}} \right)
\end{aligned}$$

$$\begin{aligned}
&= \log \left(\frac{E \left\{ \frac{P(Y=0 | U, X)P(Y(1)=1 | U, X)}{P(Y=0 | A=1, U, X)} \mid X \right\}}{E \left\{ \frac{P(Y=0 | U, X)P(Y(0)=1 | U, X)}{P(Y=0 | A=0, U, X)} \mid X \right\}} \right) \\
&\stackrel{A.S6}{<} \log \left(\frac{E \left\{ \frac{P(Y(1)=1 | U, X)}{1-\delta} \mid X \right\}}{E \{ (1-\delta)P(Y(0)=1 | U, X) \mid X \}} \right) \\
&= \log \left(\frac{E \{ P(Y(1)=1 | U, X) \mid X \}}{E \{ P(Y(0)=1 | U, X) \mid X \}} \right) - 2 \log(1-\delta) \\
&= \log \left(\frac{P(Y(1)=1 | X)}{P(Y(0)=1 | X)} \right) - 2 \log(1-\delta)
\end{aligned}$$

and similarly

$$\beta_0(X) > \log \left(\frac{P(Y(1)=1 | X)}{P(Y(0)=1 | X)} \right) + 2 \log(1-\delta).$$

When $\delta \approx 0$, we have

$$\beta_0(X) \approx \log \left(\frac{P(Y(1)=1 | X)}{P(Y(0)=1 | X)} \right).$$

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